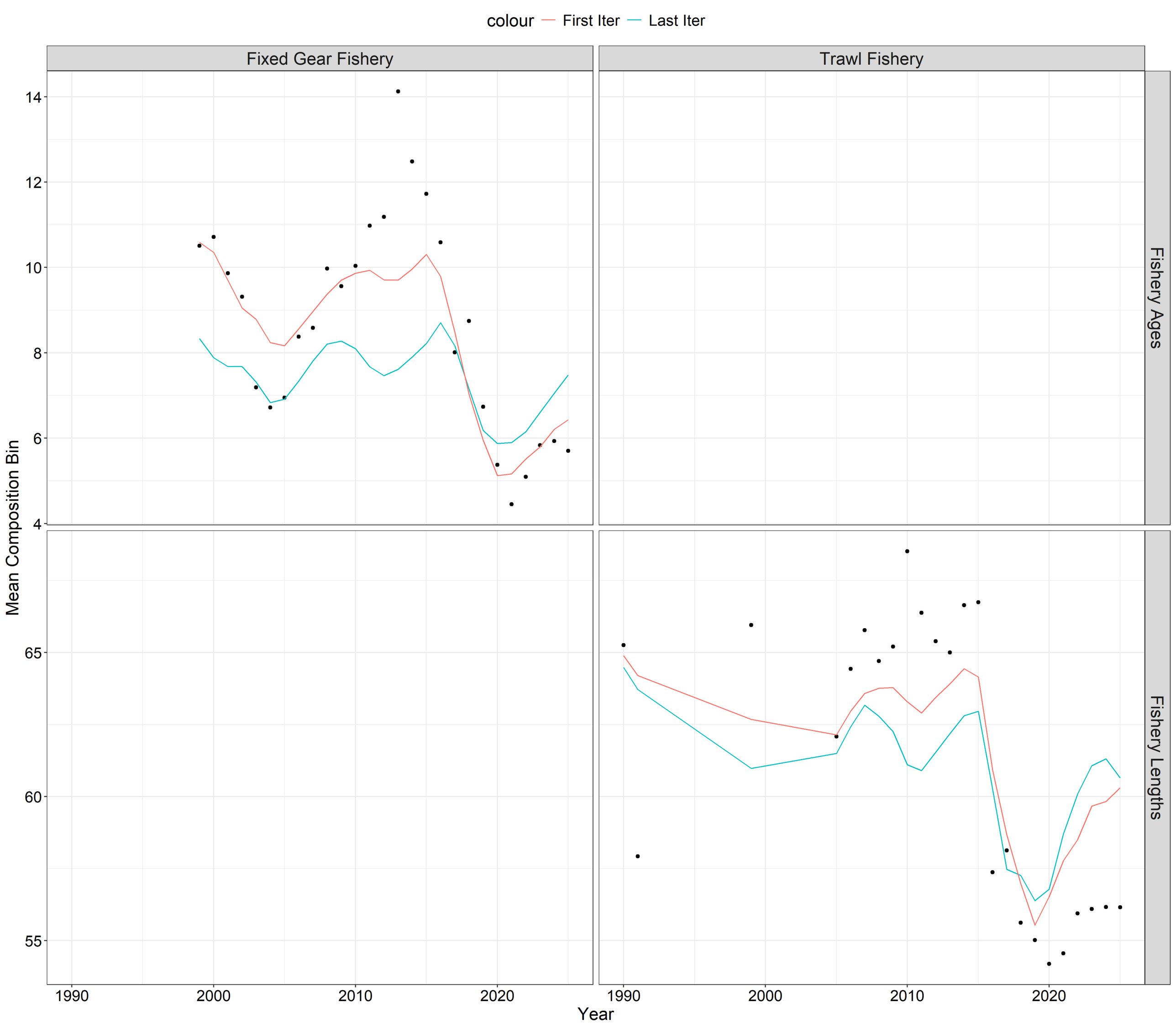

colour
First Iter
Last Iter
Fixed Gear Fishery
Trawl Fishery
Fishery Ages
Fishery Lengths
Mean Composition Bin
14
12
10
8
6
4
65
60
55
1990
2000
2010
2020
Year

colour — First Iter — Last Iter

Japan LL Survey | NOAA LL AI Survey | NOAA LL BS Survey | NOAA LL GOA Survey | NOAA Trawl Survey

Survey Ages

Survey Lengths

Mean Composition Bin

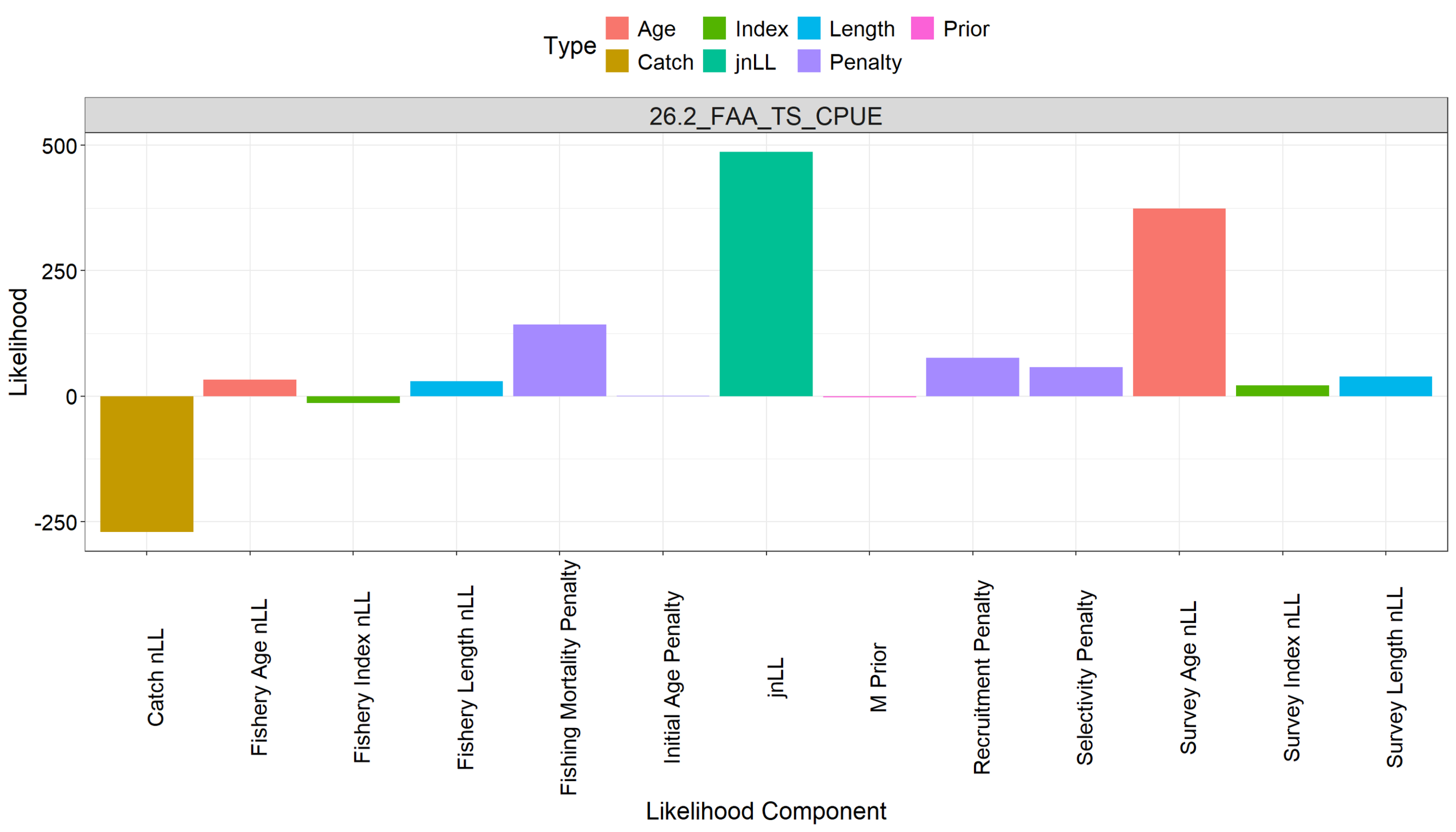
10.0
7.5
5.0

65
60
55

1980 1990 2000 2010 2020

Year


AK
JPN LLS Lengths
TS Lengths
JPN LLS Index
TS Index
GOA LLS Index
AI LLS Index
BS LLS Index
JPN LLS Ages
GOA LLS Ages
AI LLS Ages
BS LLS Ages
TF Lengths
LLF Index
TF Catch
LLF Catch
LLF Ages
26.2_FAA_TS_CPUE
1960
1980
2000
2020
Year


Type
Age
Index
Length
Prior
Catch
jnLL
Penalty
26.2_FAA_TS_CPUE
500
250
0
-250
Likelihood
Catch nLL
Fishery Age nLL
Fishery Index nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
M Prior
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Survey Length nLL
Likelihood Component

Numbers-at-Age

Sex — Female ····· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

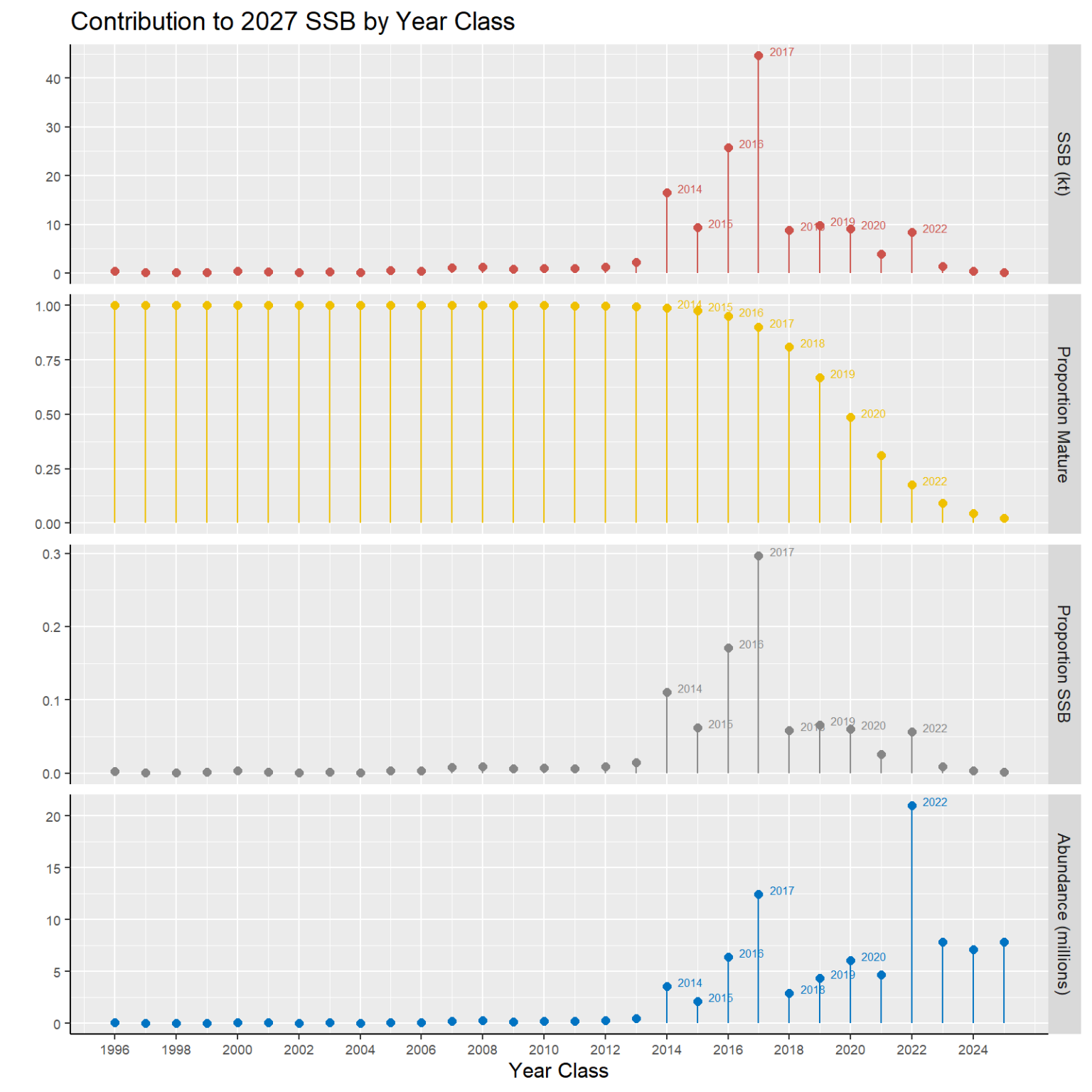
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

Numbers-at-Age (millions)

Year


Contribution to 2027 SSB by Year Class
SSB (kt)
Proportion Mature
Proportion SSB
Abundance (millions)
Year Class
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024

Model 26.2_FAA_TS_CPUE

AK

Dynamic SSB0

Recruitment

SSB

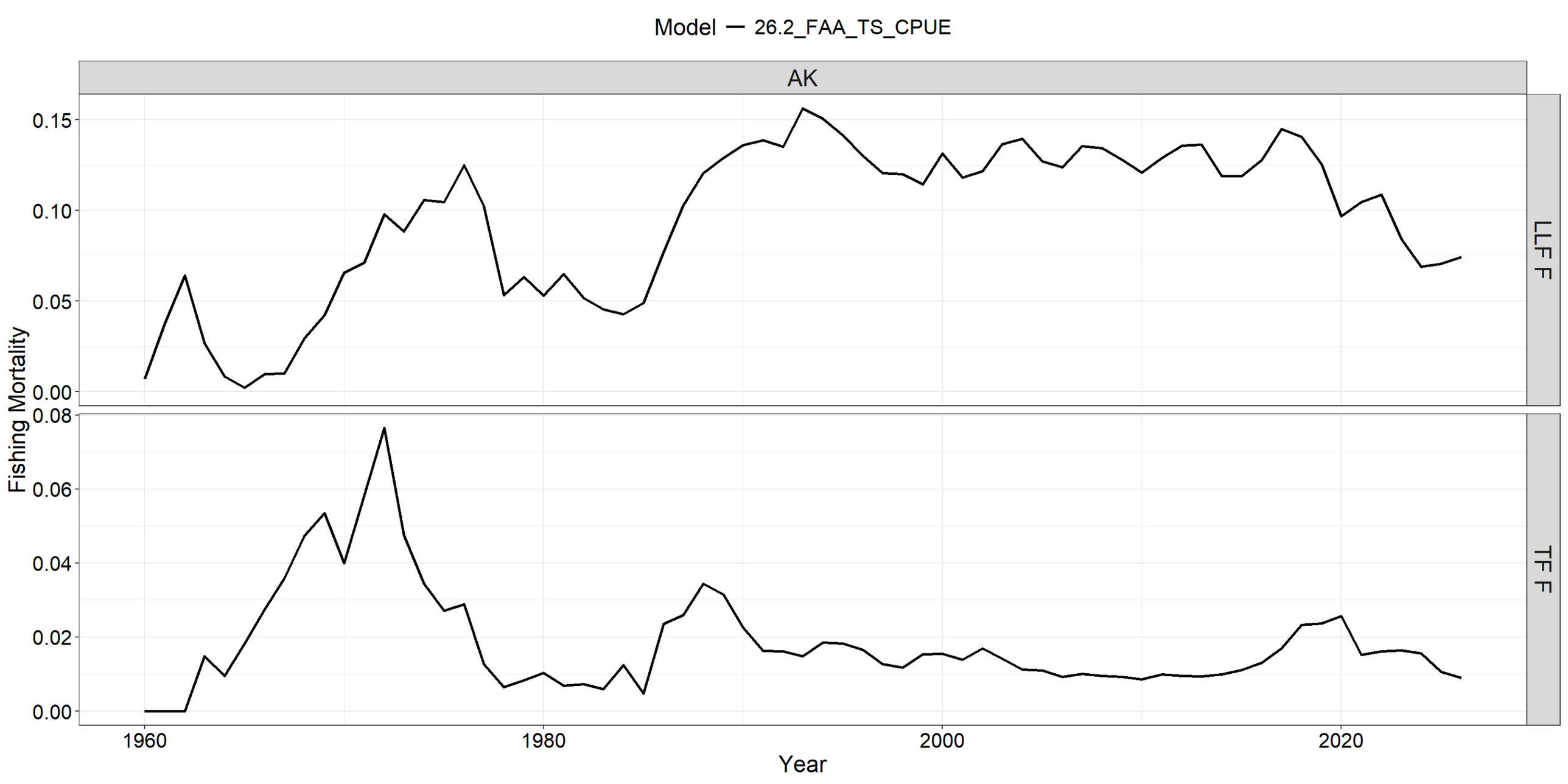
Total Biom

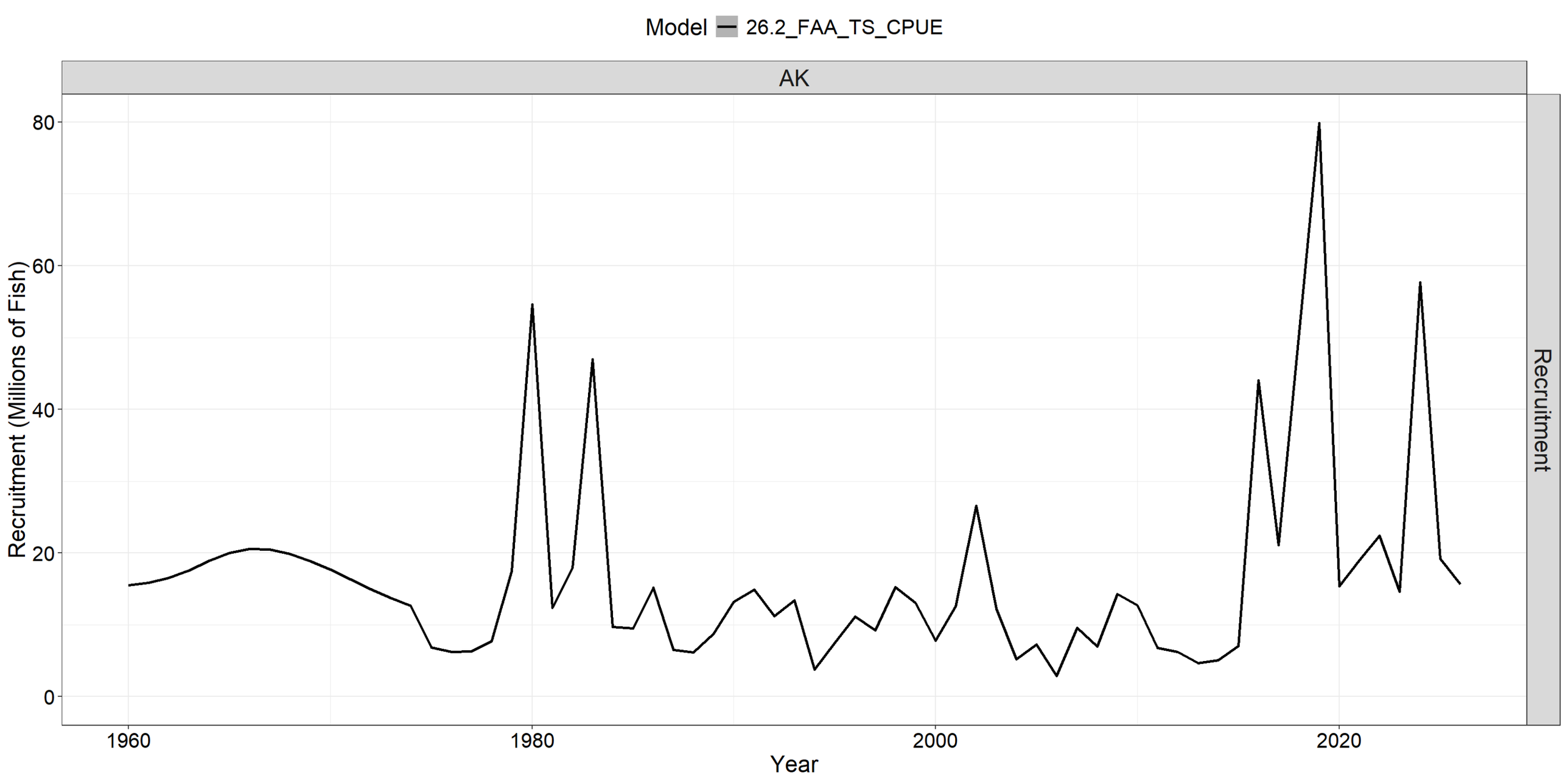
LLF F

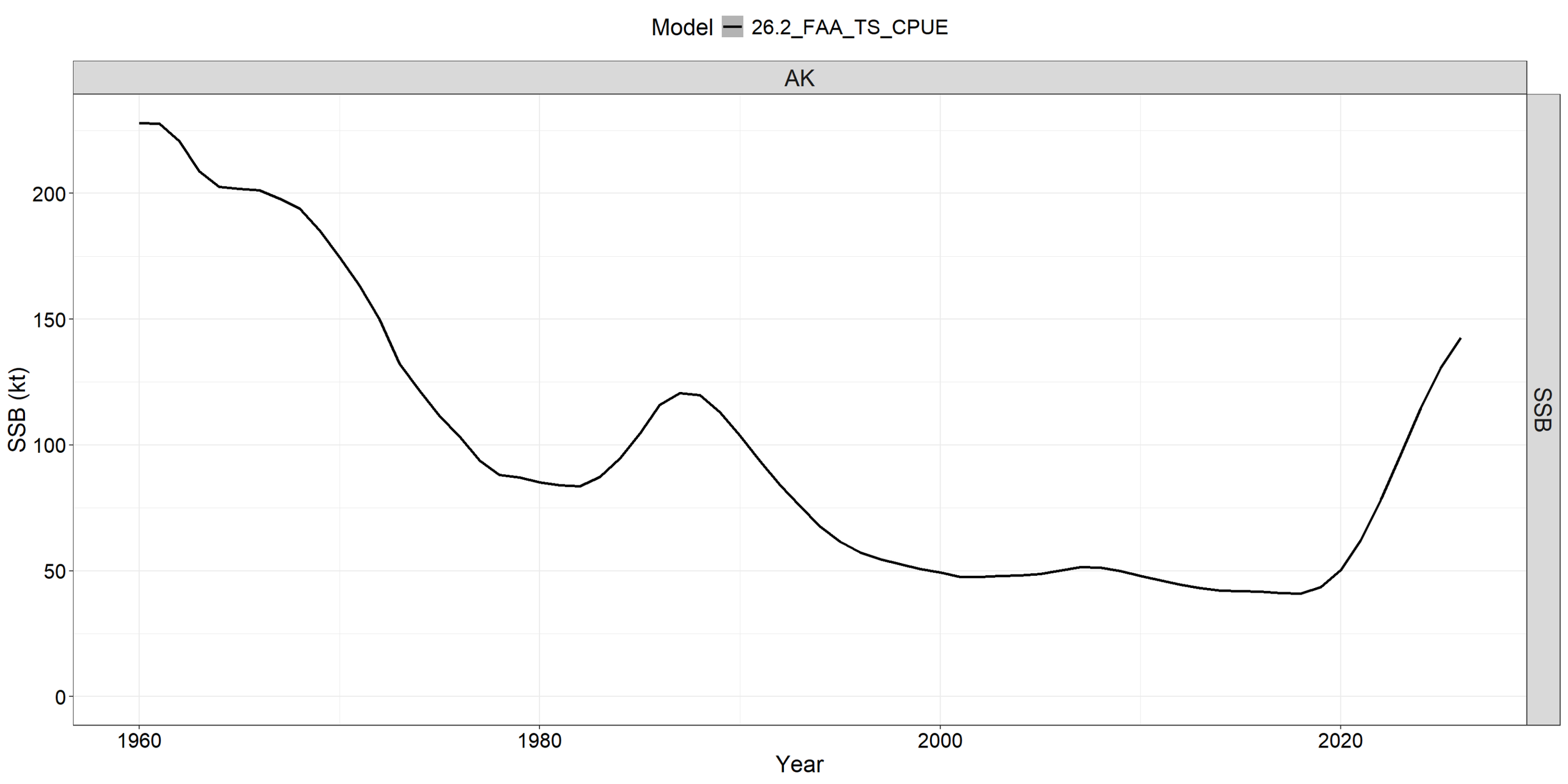
TF F

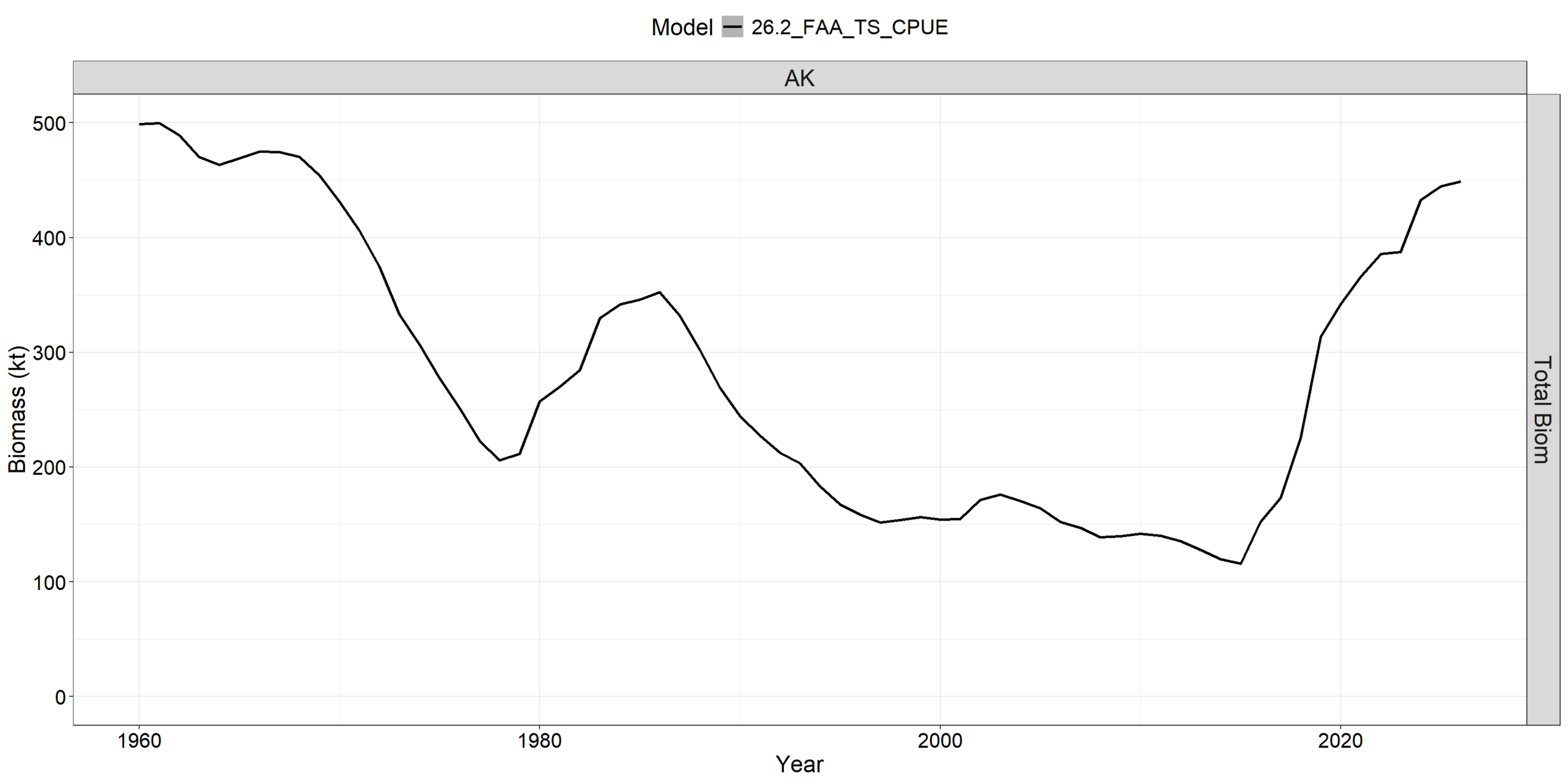
Value

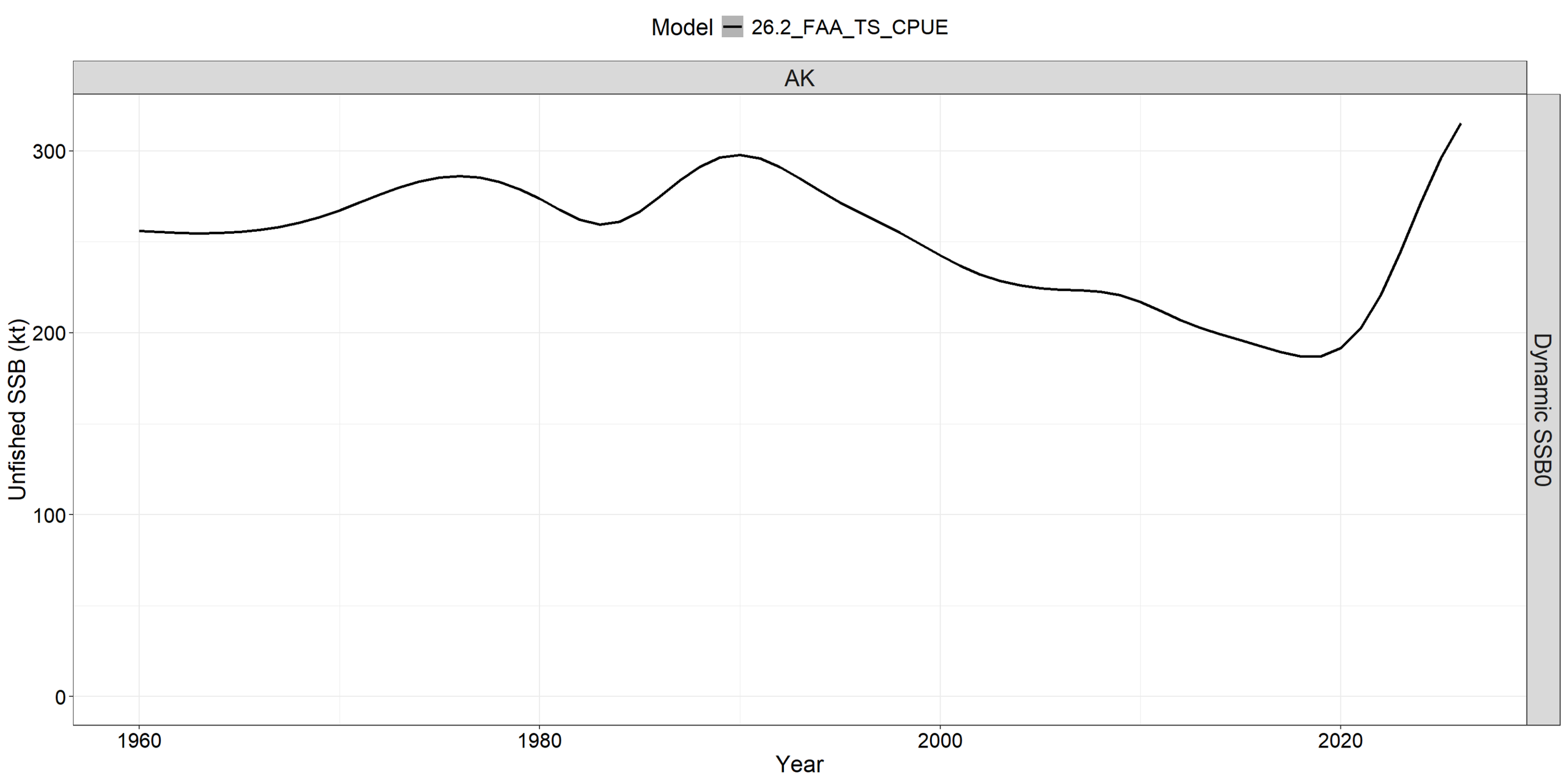
Year

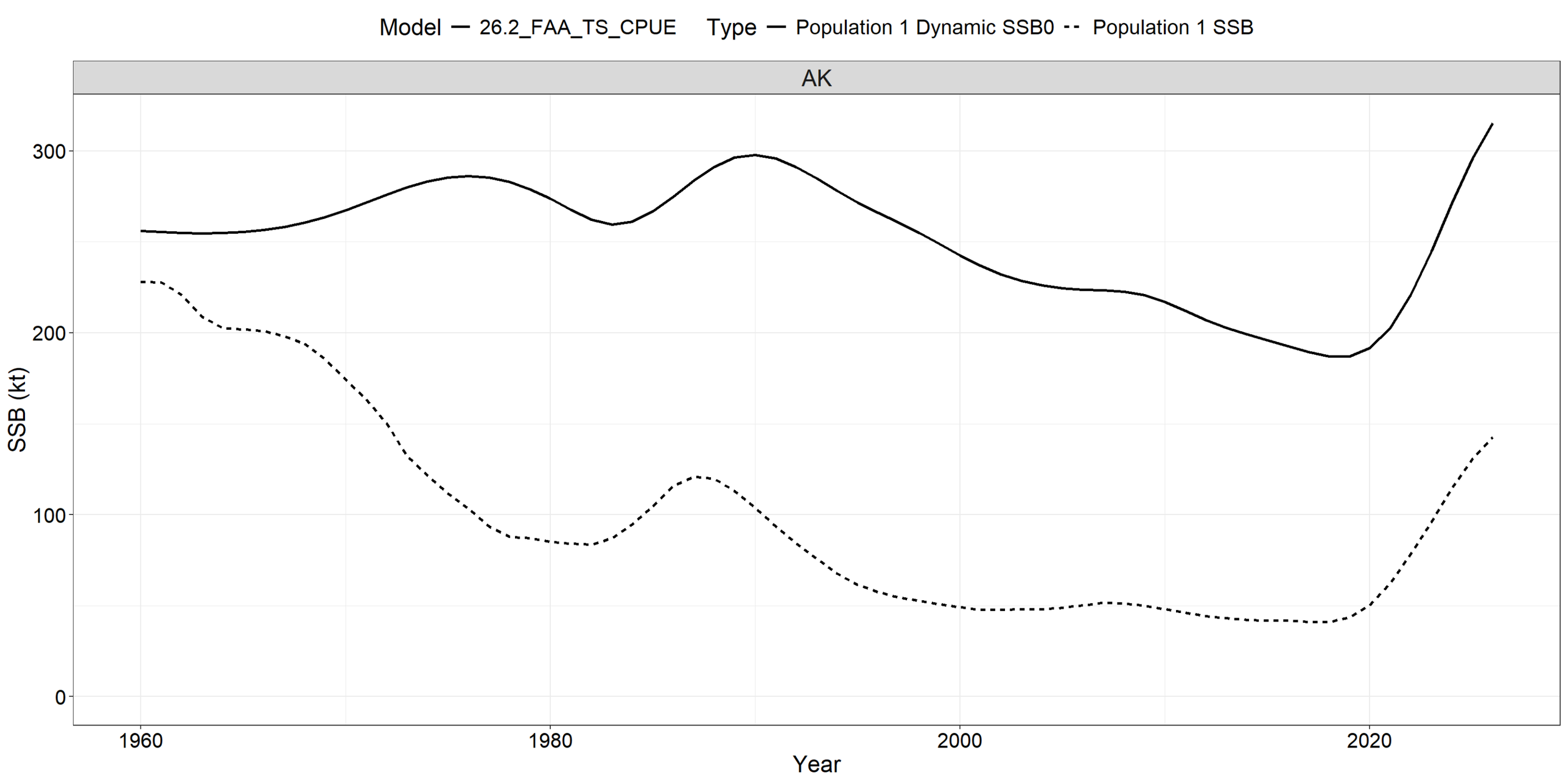

Model — 26.2_FAA_TS_CPUE
AK
LLF F
TF F
Fishing Mortality
0.15
0.10
0.05
0.00
0.08
0.06
0.04
0.02
0.00
1960
1980
2000
2020
Year

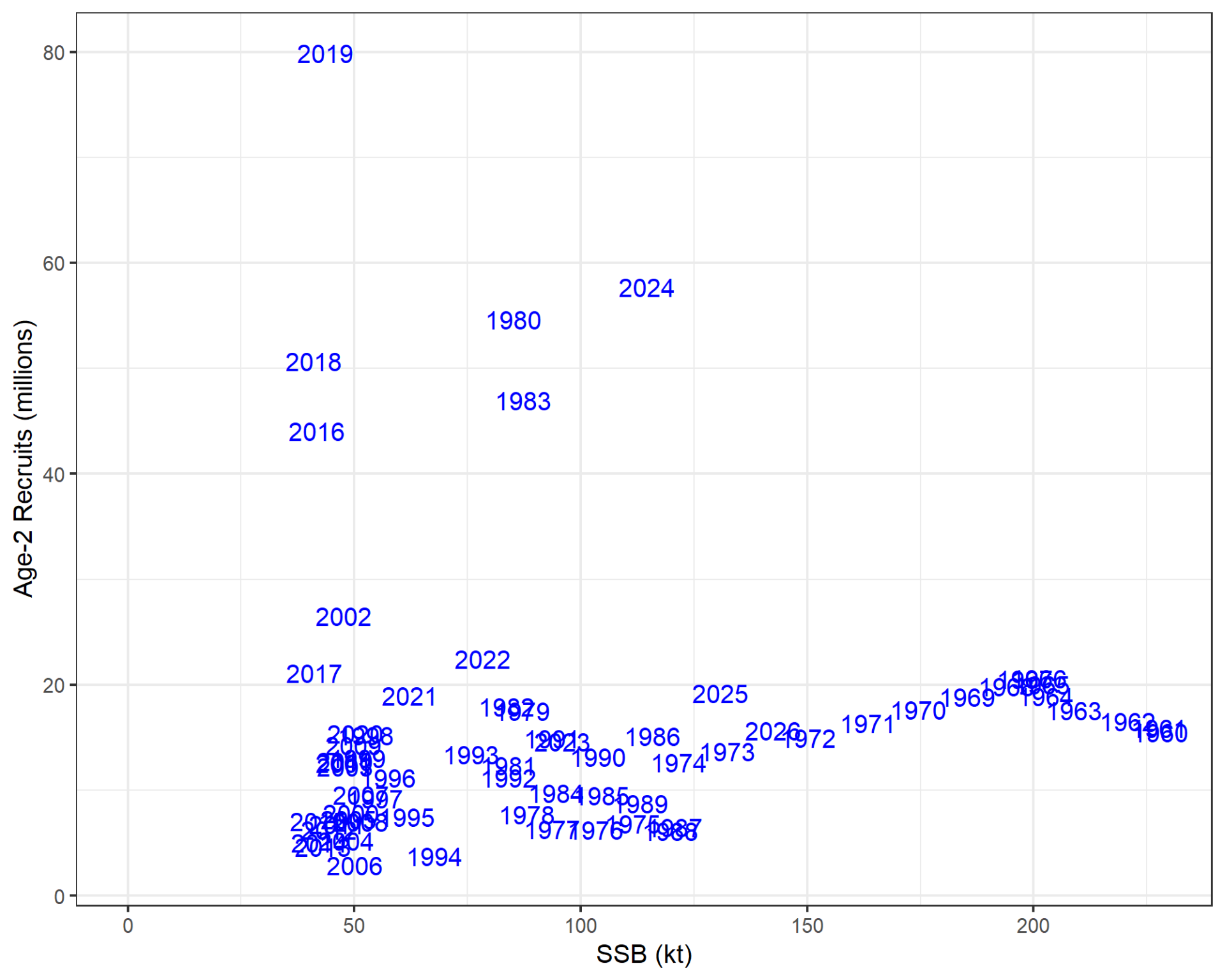

Model
26.2_FAA_TS_CPUE
AK
Recruitment (Millions of Fish)
0
20
40
60
80
1960
1980
2000
2020
Year
Recruitment

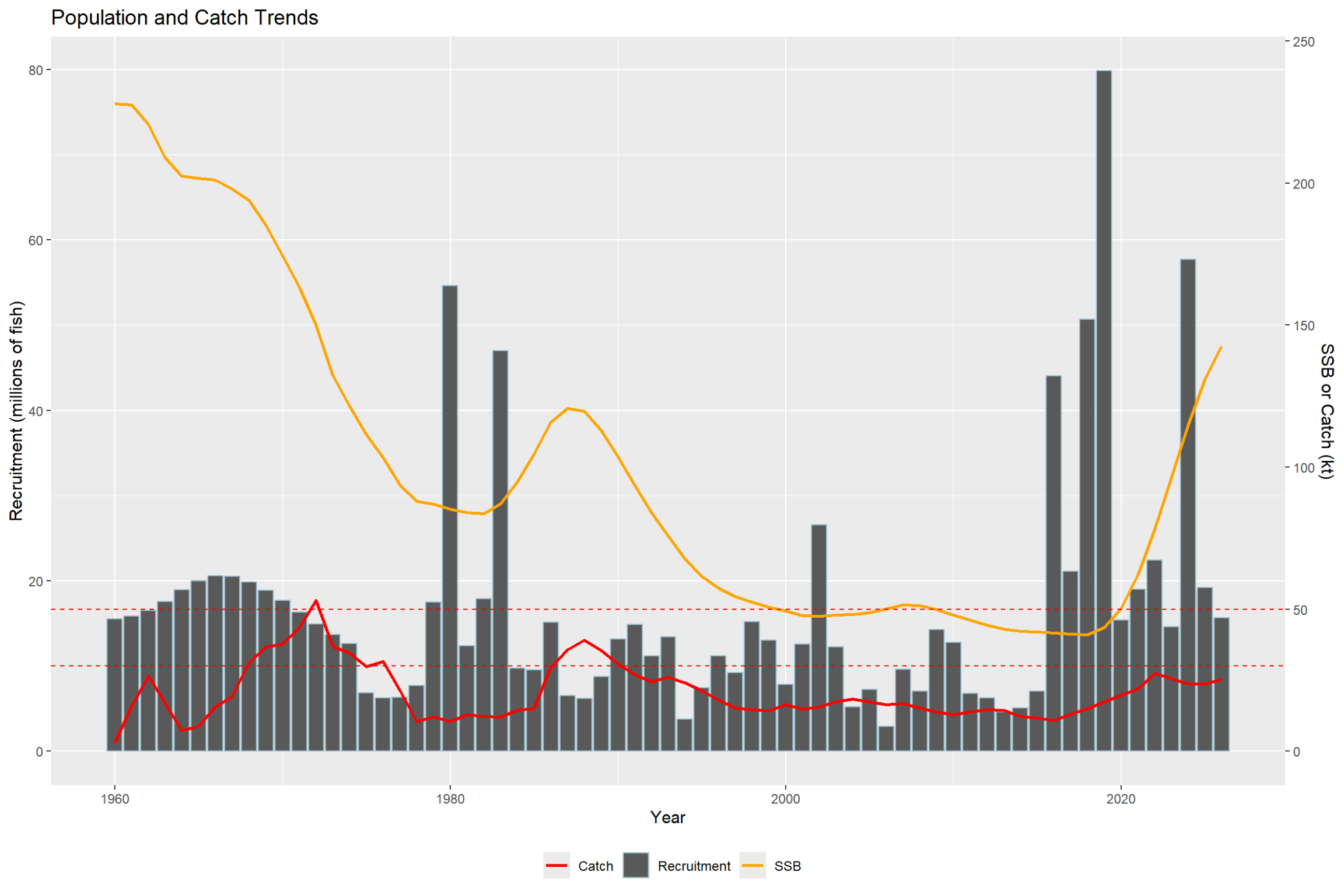

Model 26.2_FAA_TS_CPUE
AK
SSB
SSB (kt)
200
150
100
50
0
1960
1980
2000
2020
Year

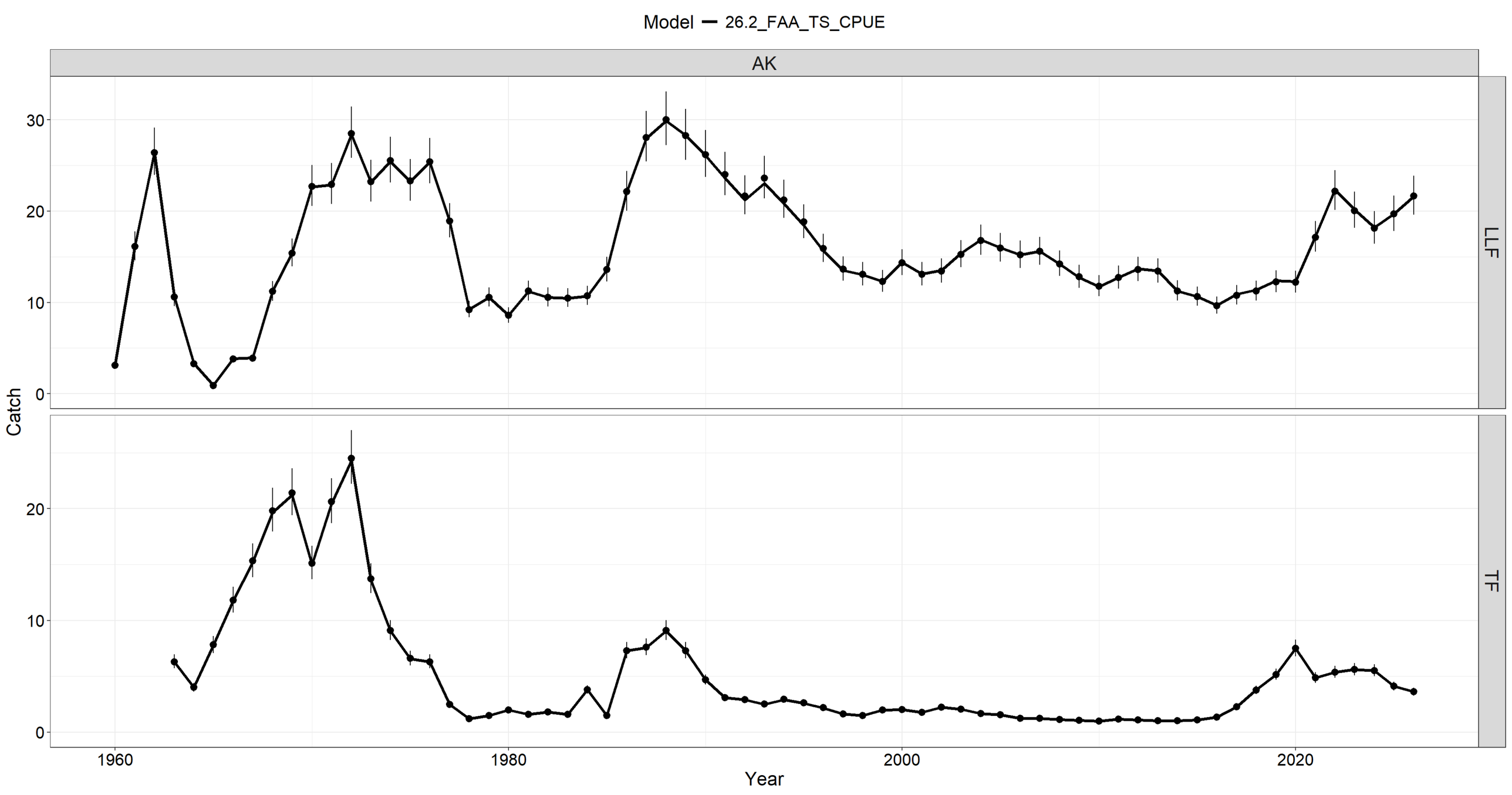

Model
26.2_FAA_TS_CPUE
AK
Total Biom
Biomass (kt)
500
400
300
200
100
0
1960
1980
2000
2020
Year


Model
26.2_FAA_TS_CPUE
AK
Dynamic SSB0
Unfished SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year


Model — 26.2_FAA_TS_CPUE Type — Population 1 Dynamic SSB0 -- Population 1 SSB
AK
300
200
100
0
SSB (kt)
1960
1980
2000
2020
Year


Age-2 Recruits (millions)
SSB (kt)
0
20
40
60
80
0
50
100
150
200
2019
2024
1980
2018
1983
2016
2002
2022
2017
2021
2025
1971
1970
1969
1963
1972
1973
1974
1990
1986
1993
1996
1994
2006
1995
1978
1989
1984
1985
1992
1981
1979


Population and Catch Trends
Recruitment (millions of fish)
SSB or Catch (kt)
Year
0
20
40
60
80
0
50
100
150
200
250
1960
1980
2000
2020
Catch
Recruitment
SSB


Model — 26.2_FAA_TS_CPUE
AK
LLF
TF
Catch
Year
0
10
20
30
1960
1980
2000
2020

Model — 26.2_FAA_TS_CPUE

LLF Q2

LLF Q3

BS LLS

AI LLS

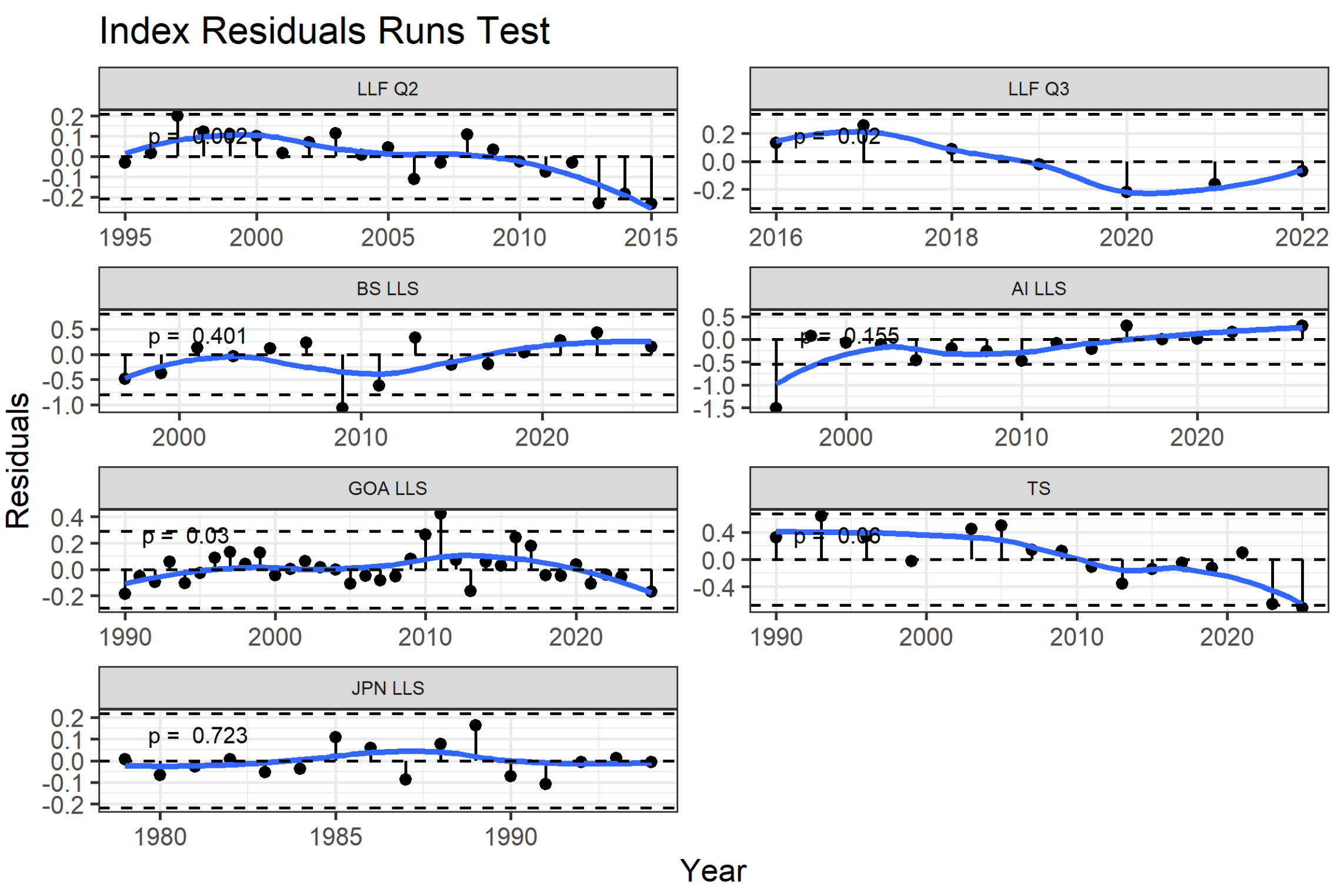
GOA LLS

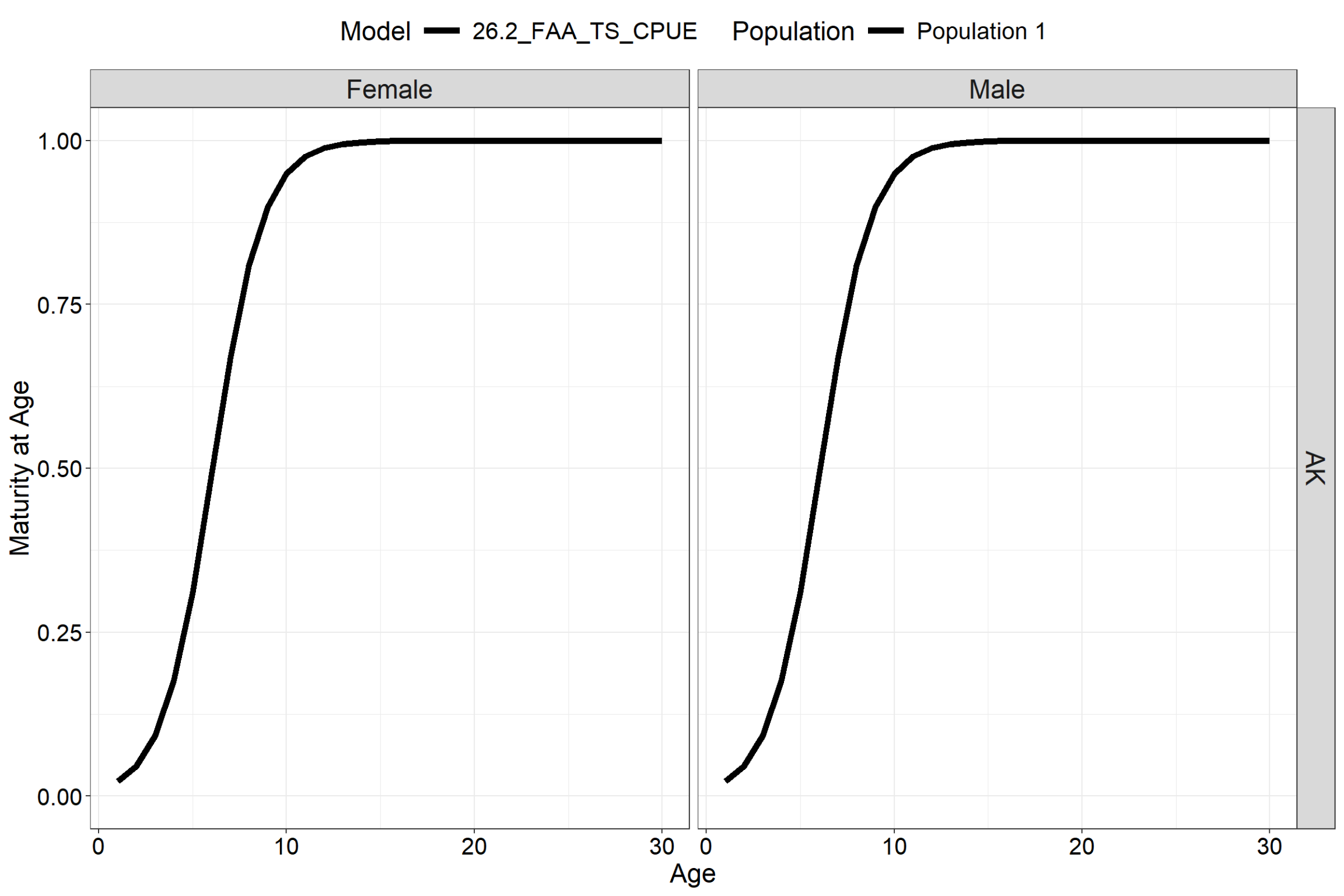
TS

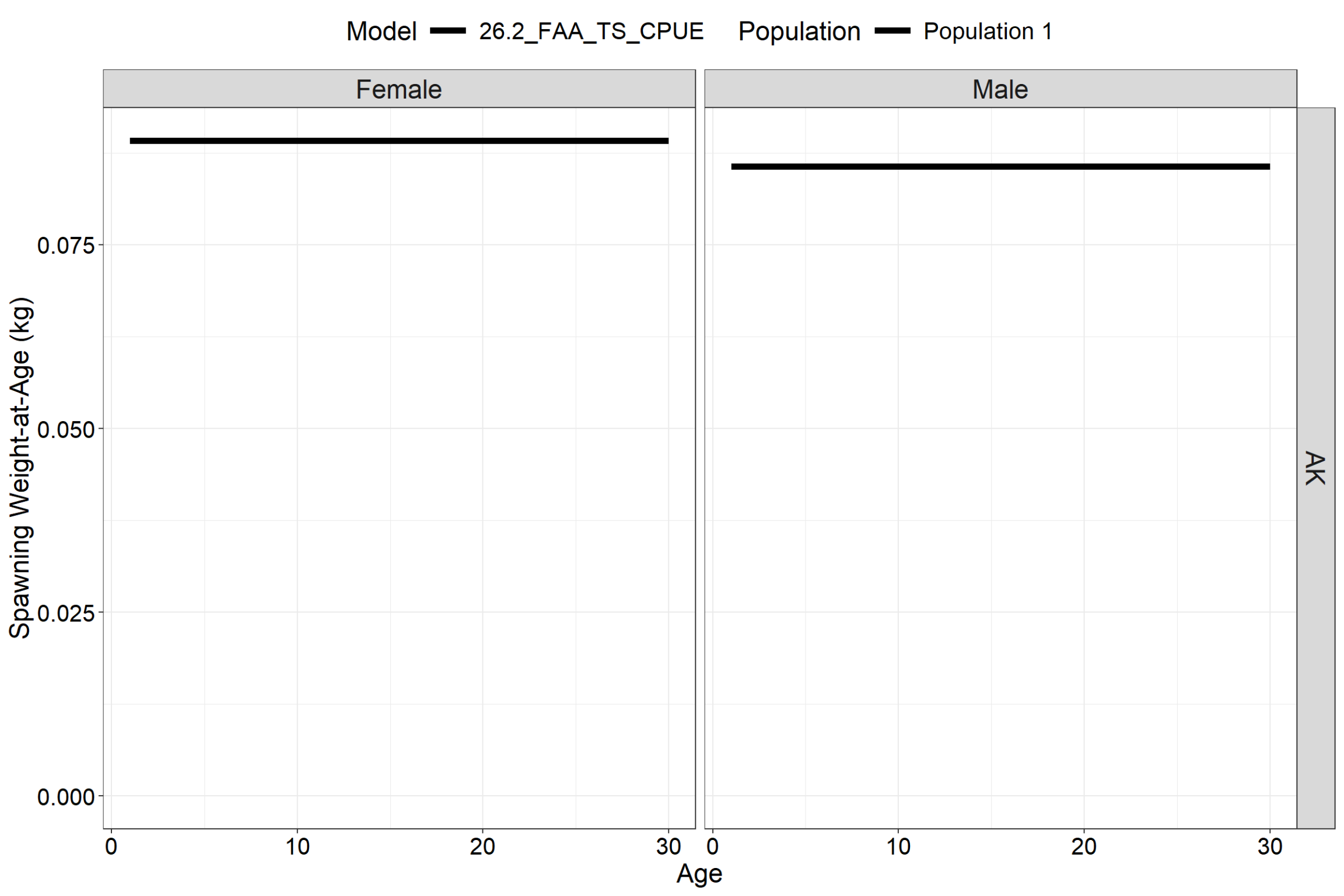
JPN LLS

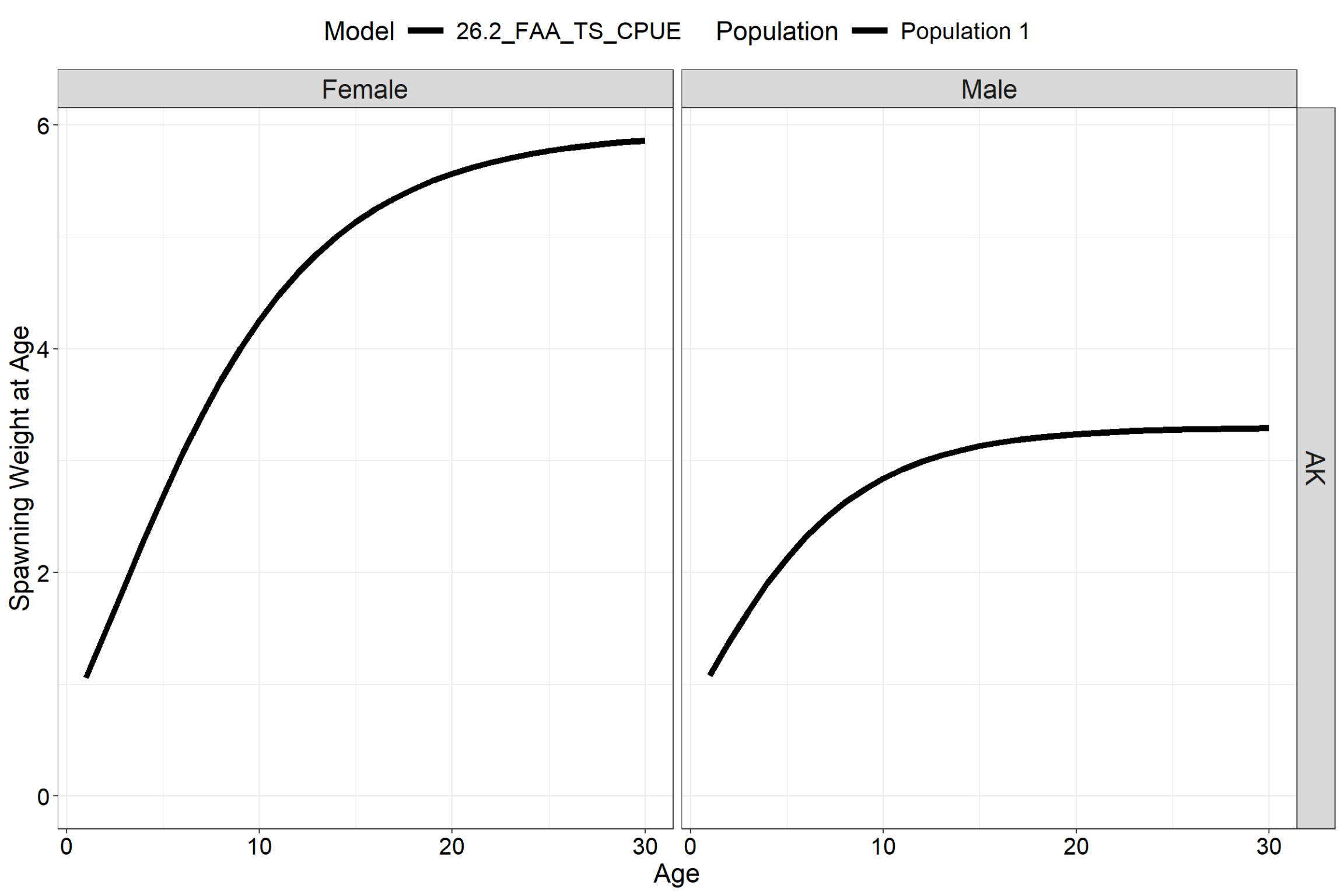
RPNs (1000s)

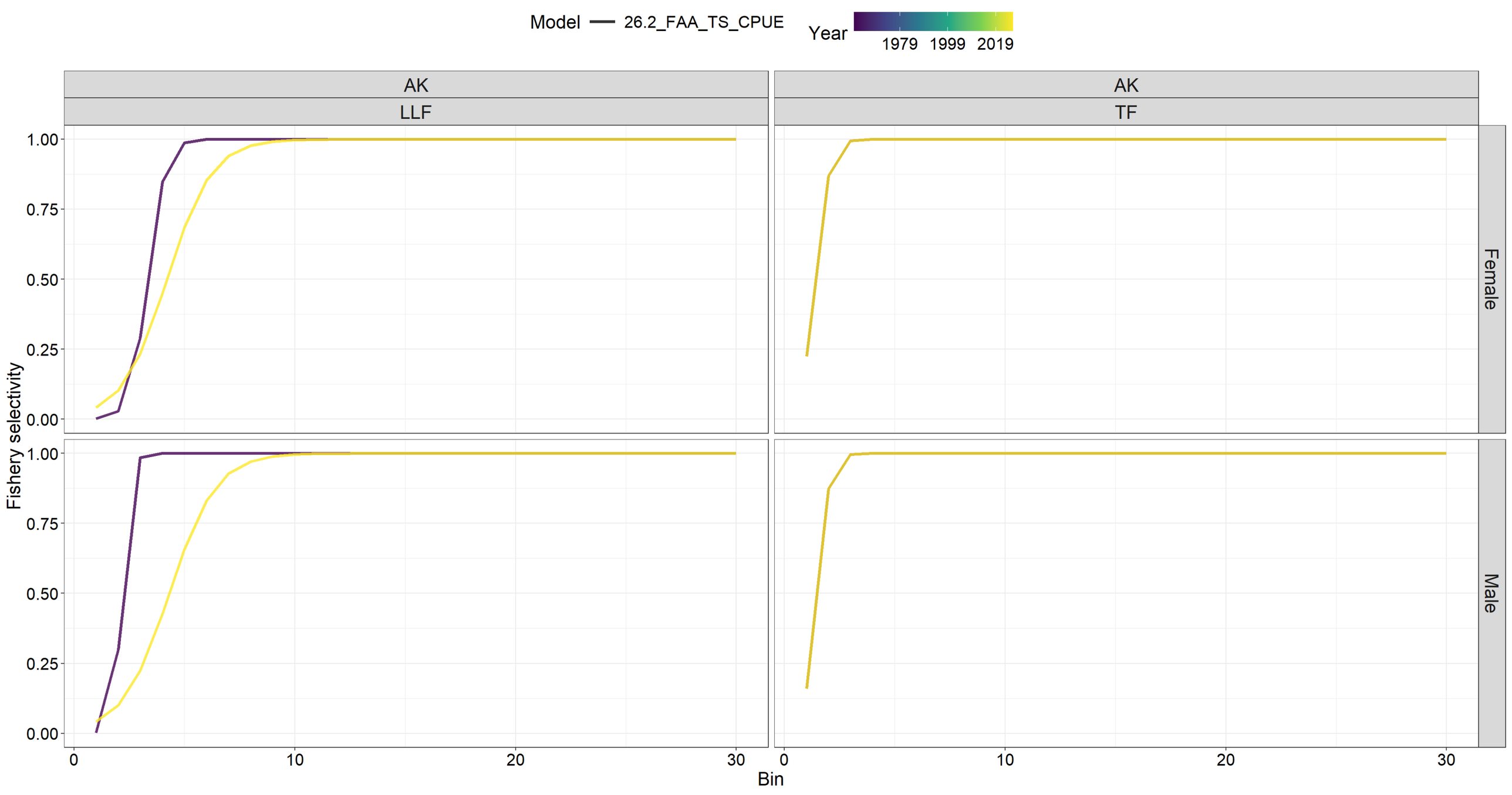
Year

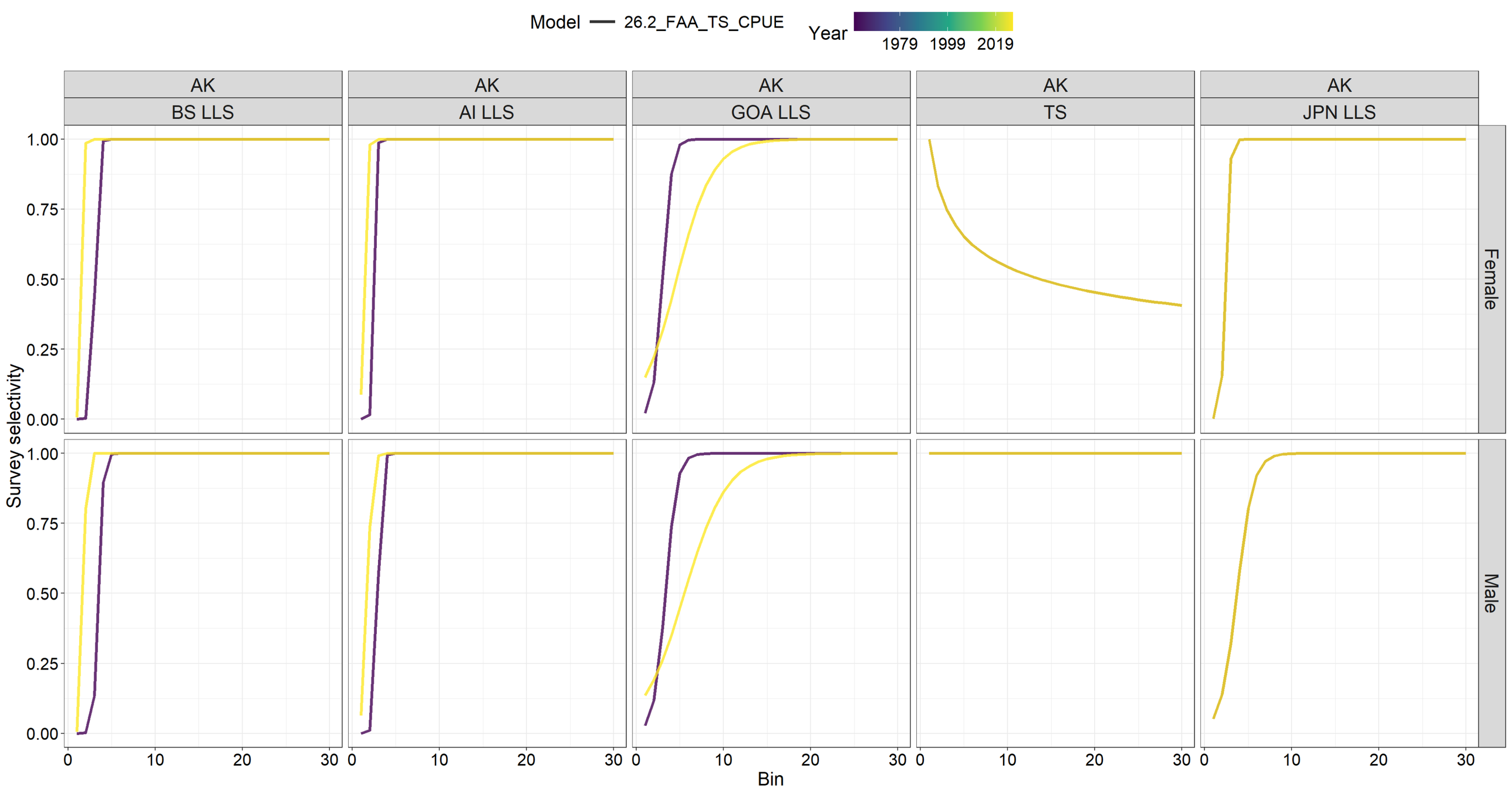

Index Residuals Runs Test
LLF Q2
p = 0.002
LLF Q3
p = 0.02
BS LLS
p = 0.401
AI LLS
p = 0.155
GOA LLS
p = 0.03
TS
p = 0.06
JPN LLS
p = 0.723
Residuals
Year


Model 26.2_FAA_TS_CPUE
Population Population 1
Female
Male
AK
Maturity at Age
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Age


Model
26.2_FAA_TS_CPUE
Population
Population 1
Female
Male
AK
Spawning Weight-at-Age (kg)
0.075
0.050
0.025
0.000
0
10
20
30
Age


Model
26.2_FAA_TS_CPUE
Population
Population 1
Female
Male
AK
Spawning Weight at Age
Age
0
2
4
6
0
10
20
30


Model
26.2_FAA_TS_CPUE
Year
1979
1999
2019
AK
LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Model
26.2_FAA_TS_CPUE
Year
1979
1999
2019
AK
BS LLS
AI LLS
GOA LLS
TS
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin

Phase Plane

1.5
1.0
0.5
0.0

$F/F_{35\%}$

0.0
0.5
1.0
1.5
2.0

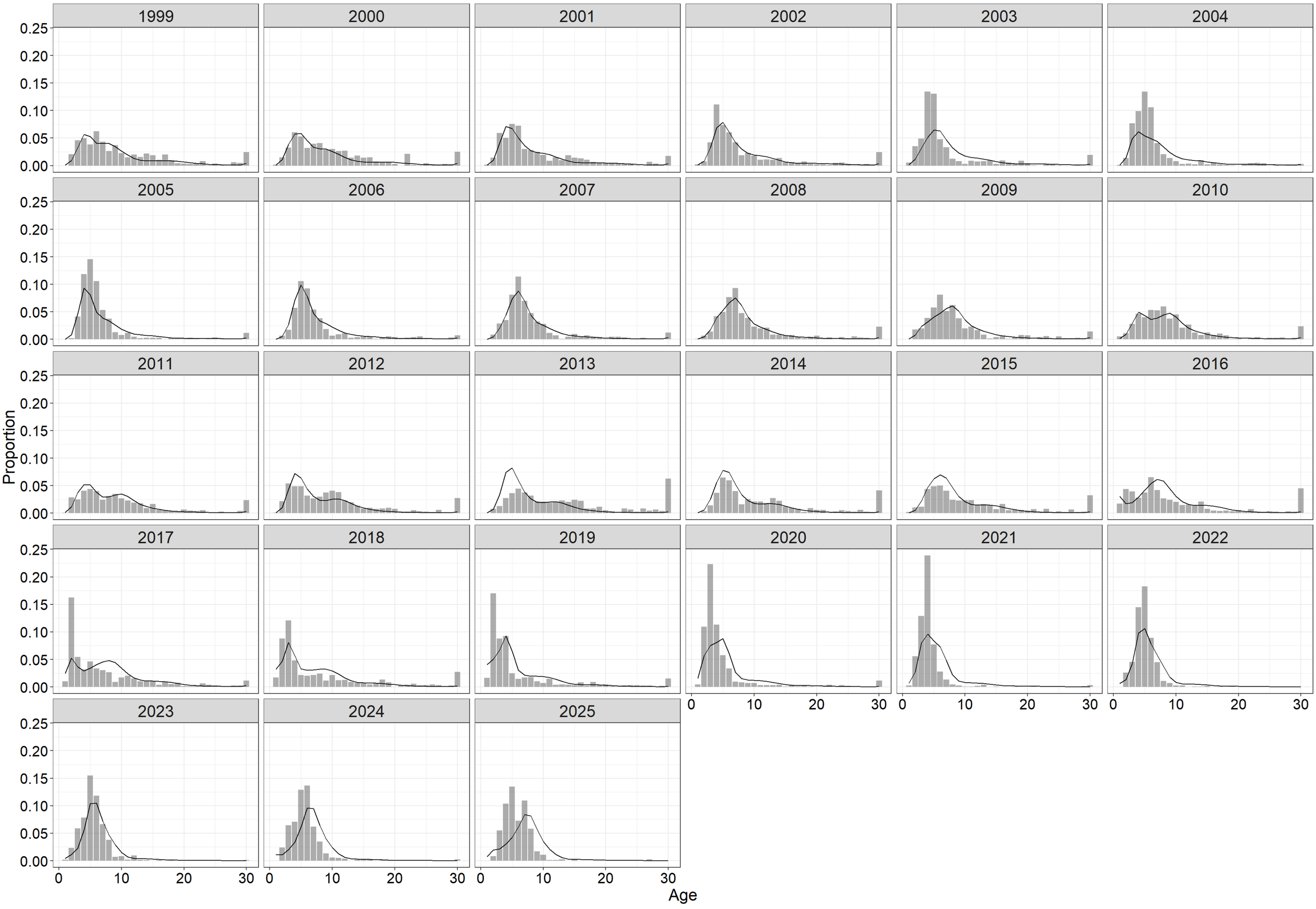
$SSB/B_{35\%}$

1995
1990
2000
2005
2015
2010
2020
1975
1970
2026
2025
1980
1985
1965
1960

Year
2020
2000
1980
1960

Control Rule
OFL (F35%)
Max ABC (F40%)

Fixed Gear Fishery Age Comps Female AK



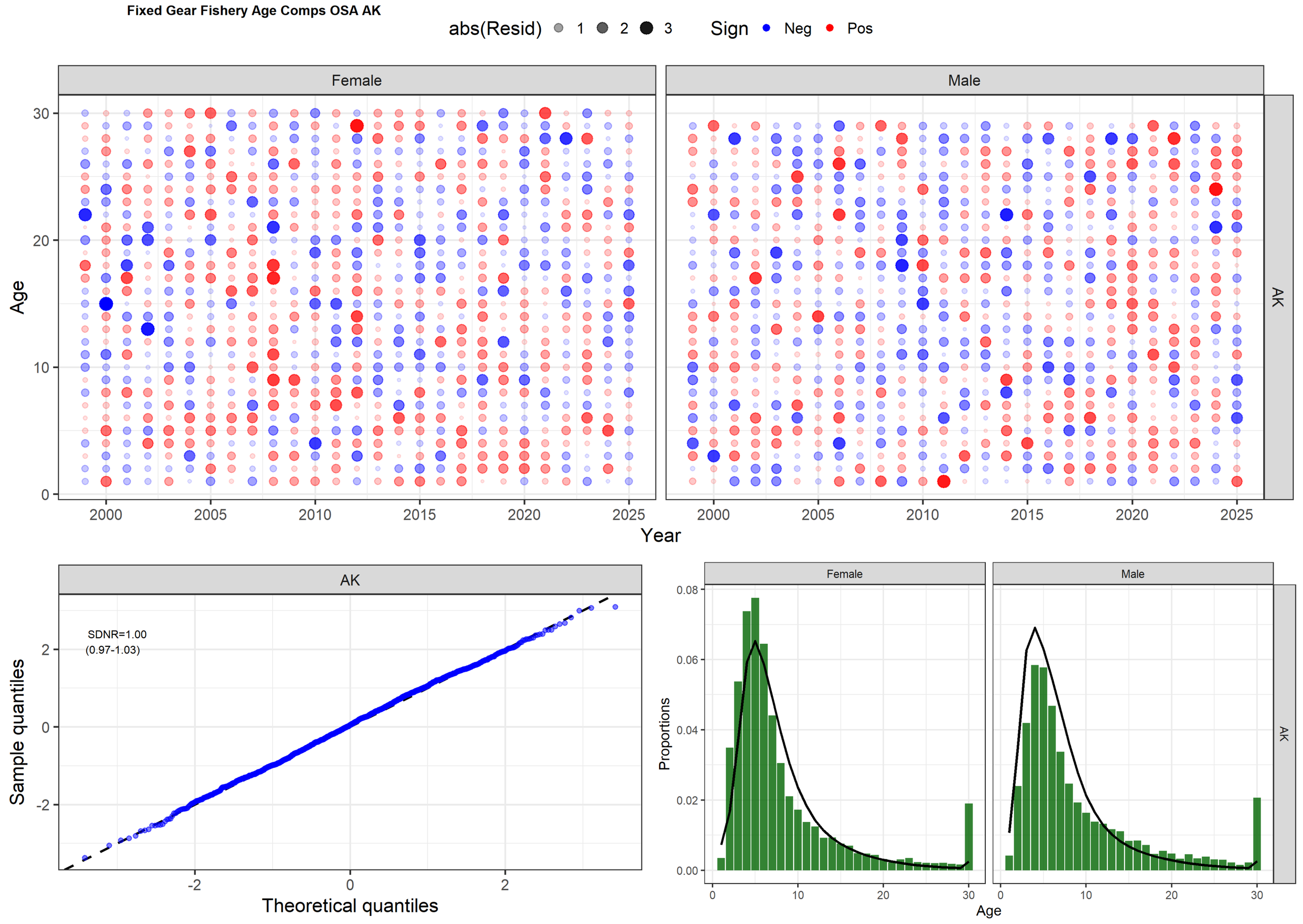
Fixed Gear Fishery Age Comps Male AK

1999 2000 2001 2002 2003 2004
2005 2006 2007 2008 2009 2010
2011 2012 2013 2014 2015 2016
2017 2018 2019 2020 2021 2022
2023 2024 2025

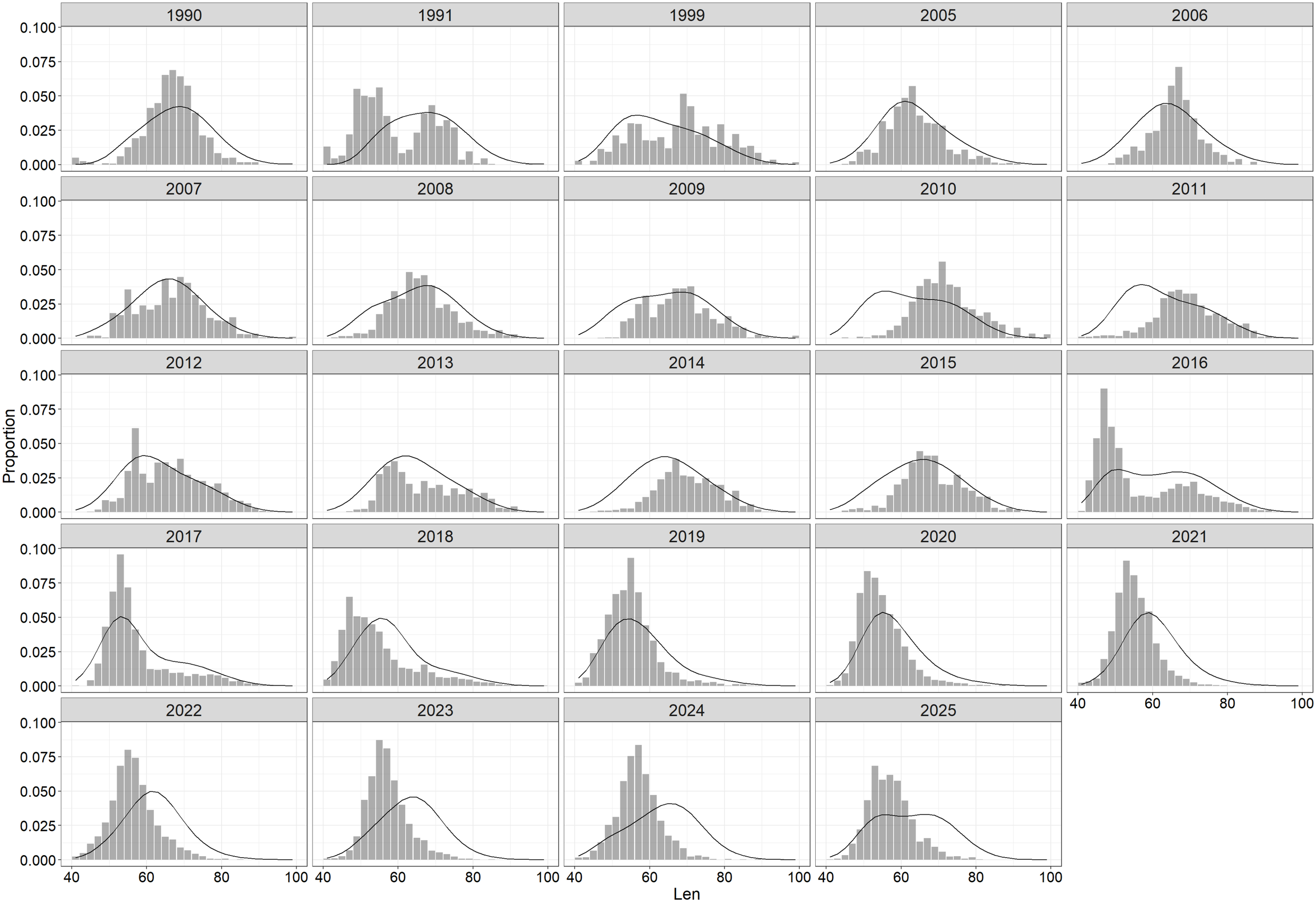
Proportion: 0.00, 0.05, 0.10, 0.15

Age: 0, 10, 20, 30

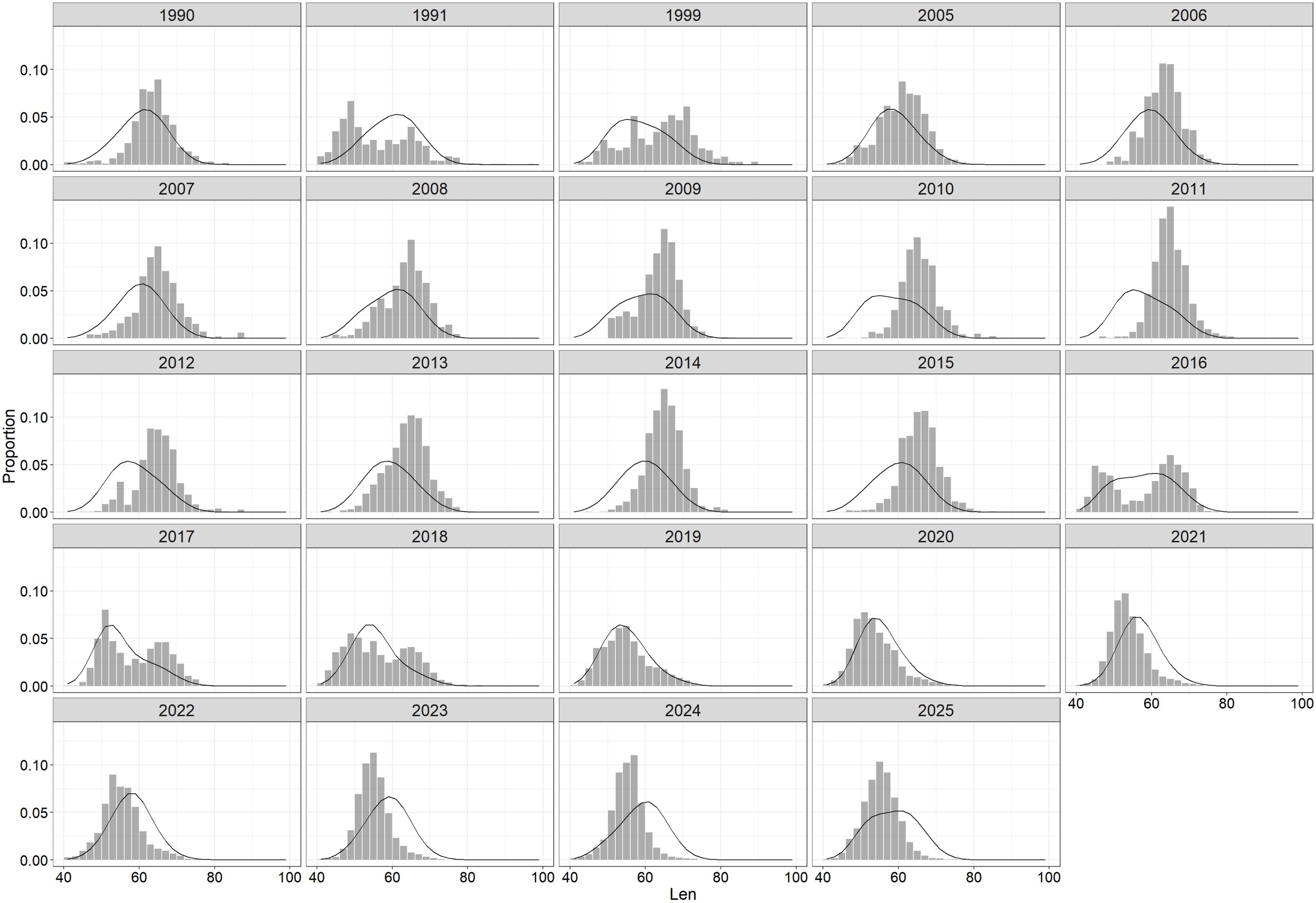
Fixed Gear Fishery Age Comps OSA AK



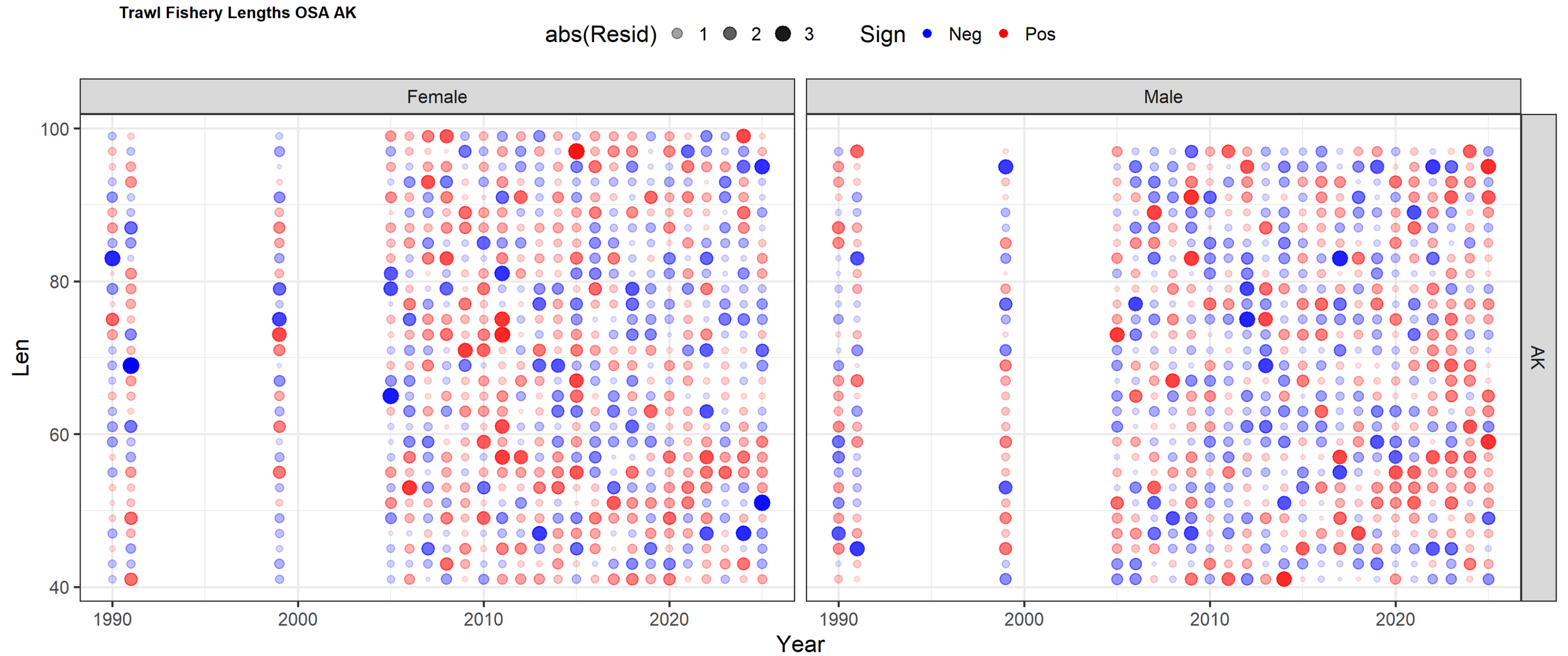
Trawl Fishery Lengths Female AK

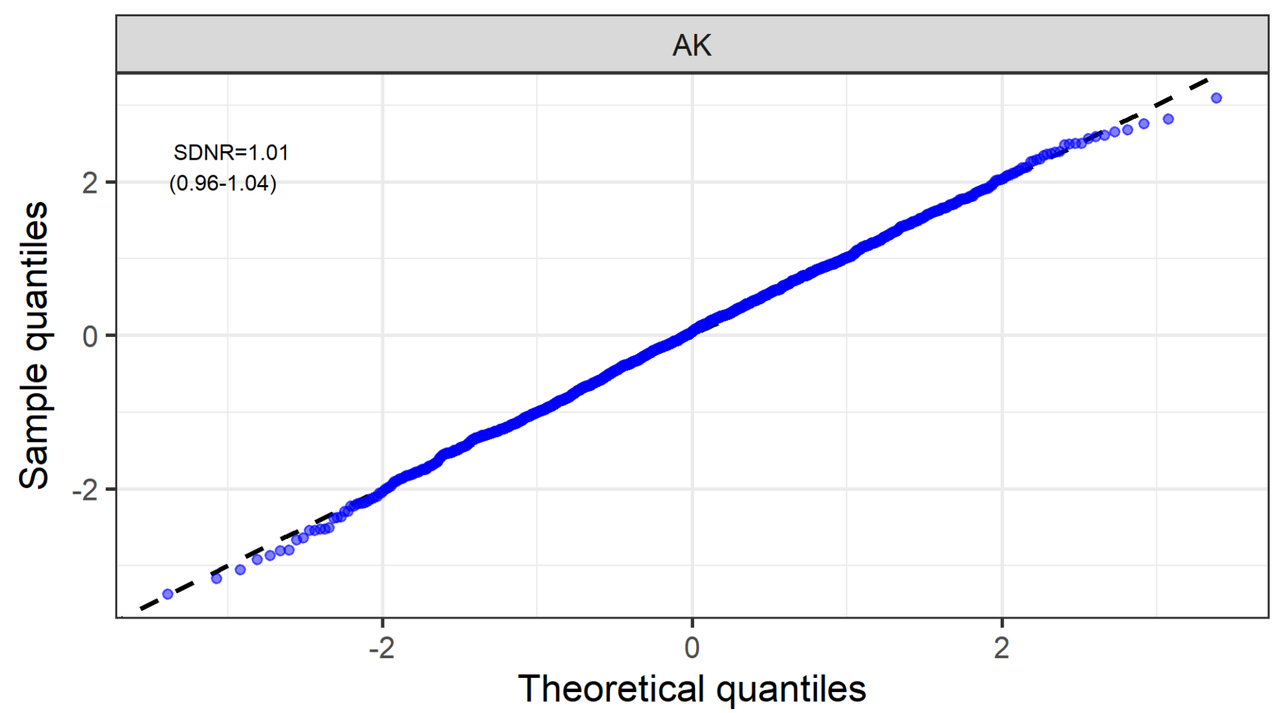


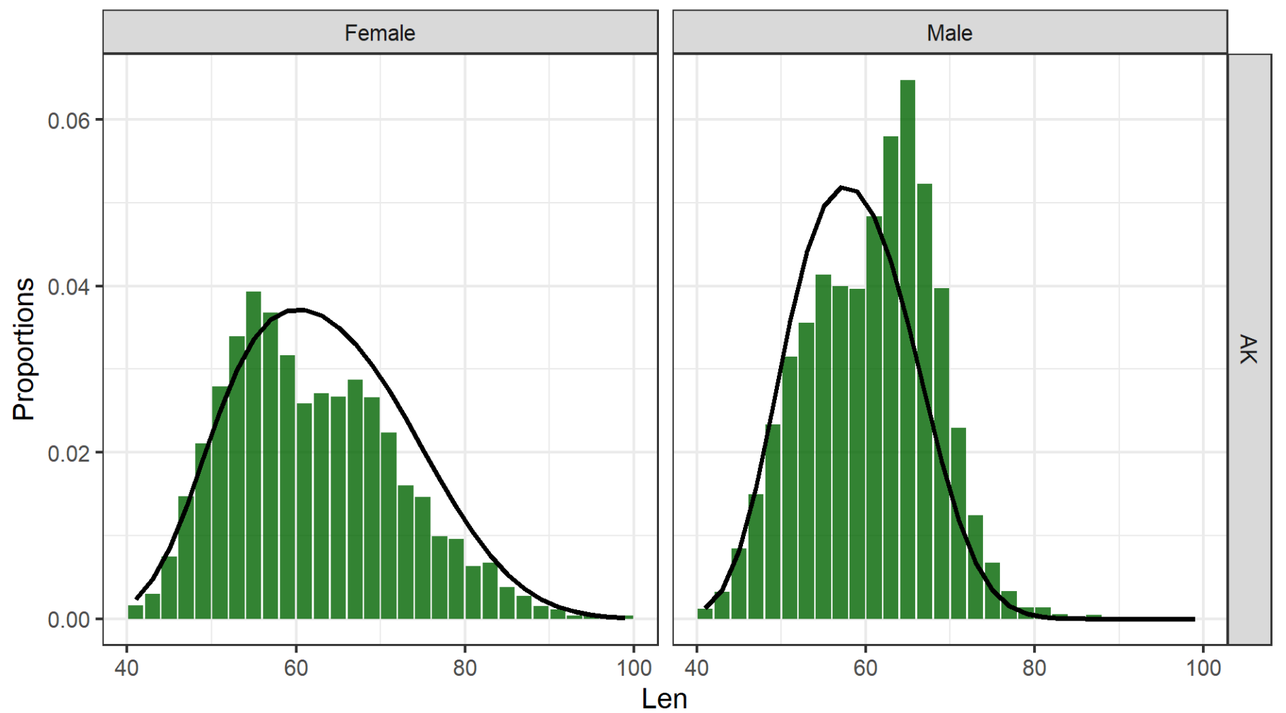
Trawl Fishery Lengths Male AK



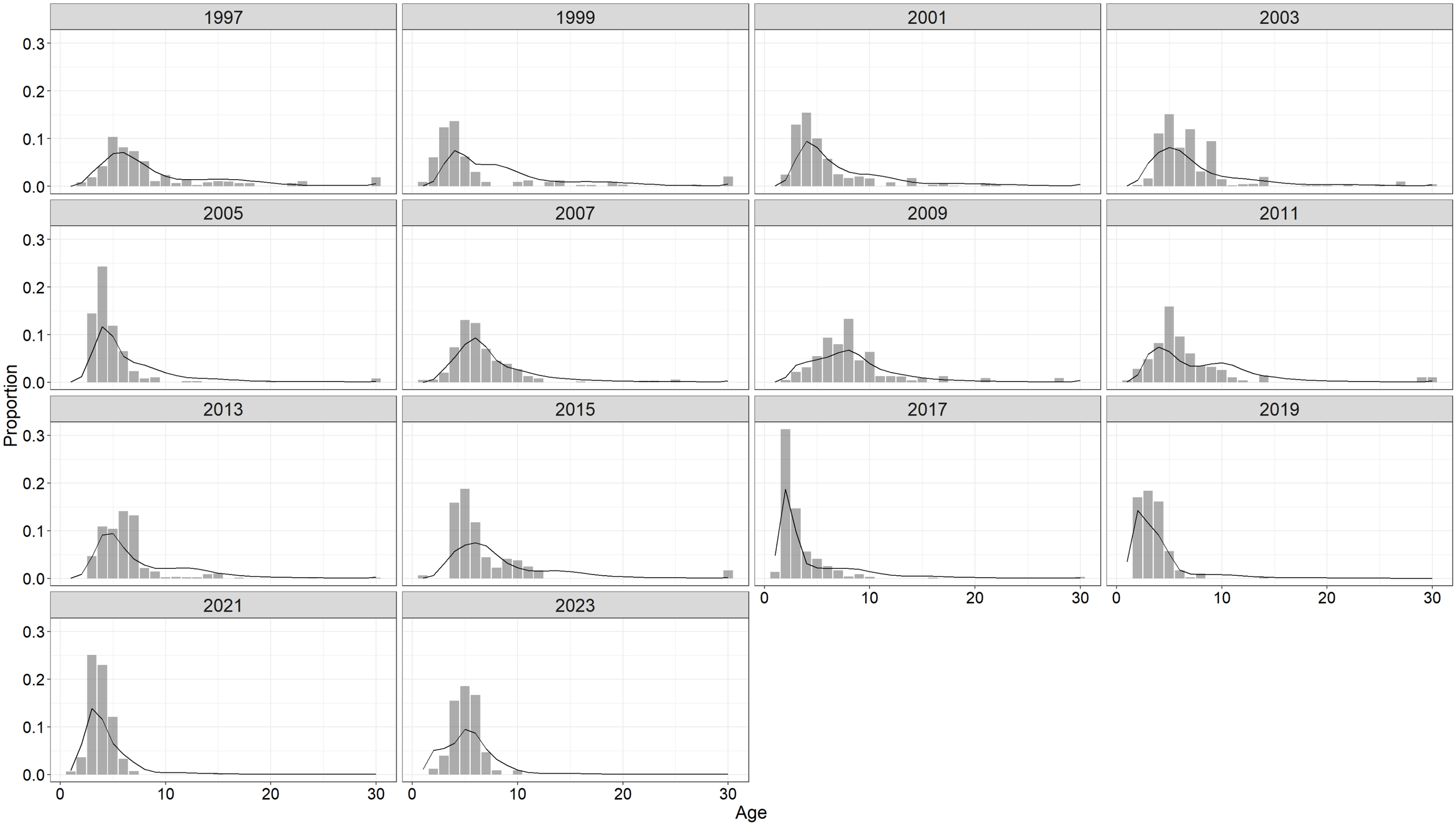
Trawl Fishery Lengths OSA AK







BS LL Survey Age Comps Female AK

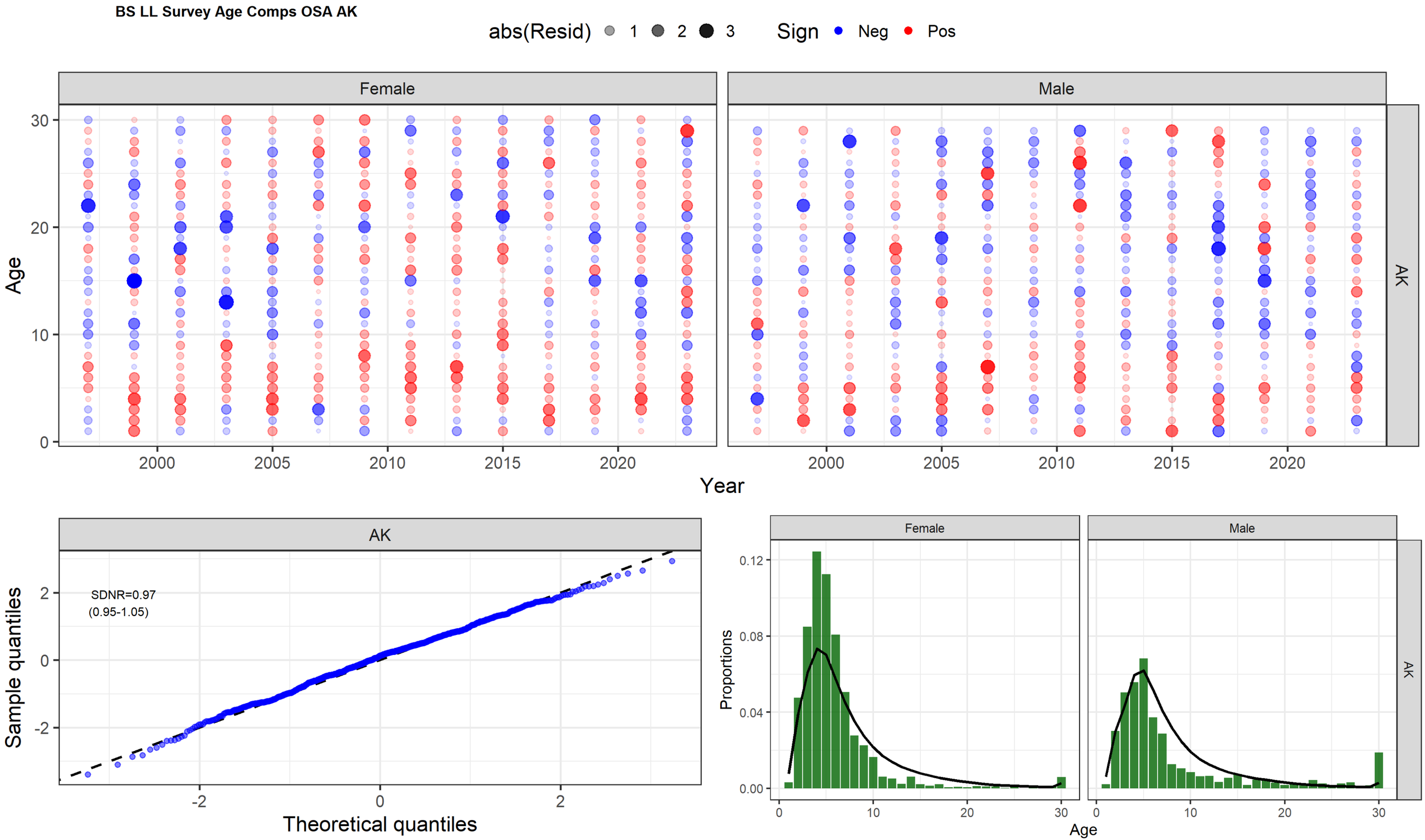


BS LL Survey Age Comps Male AK

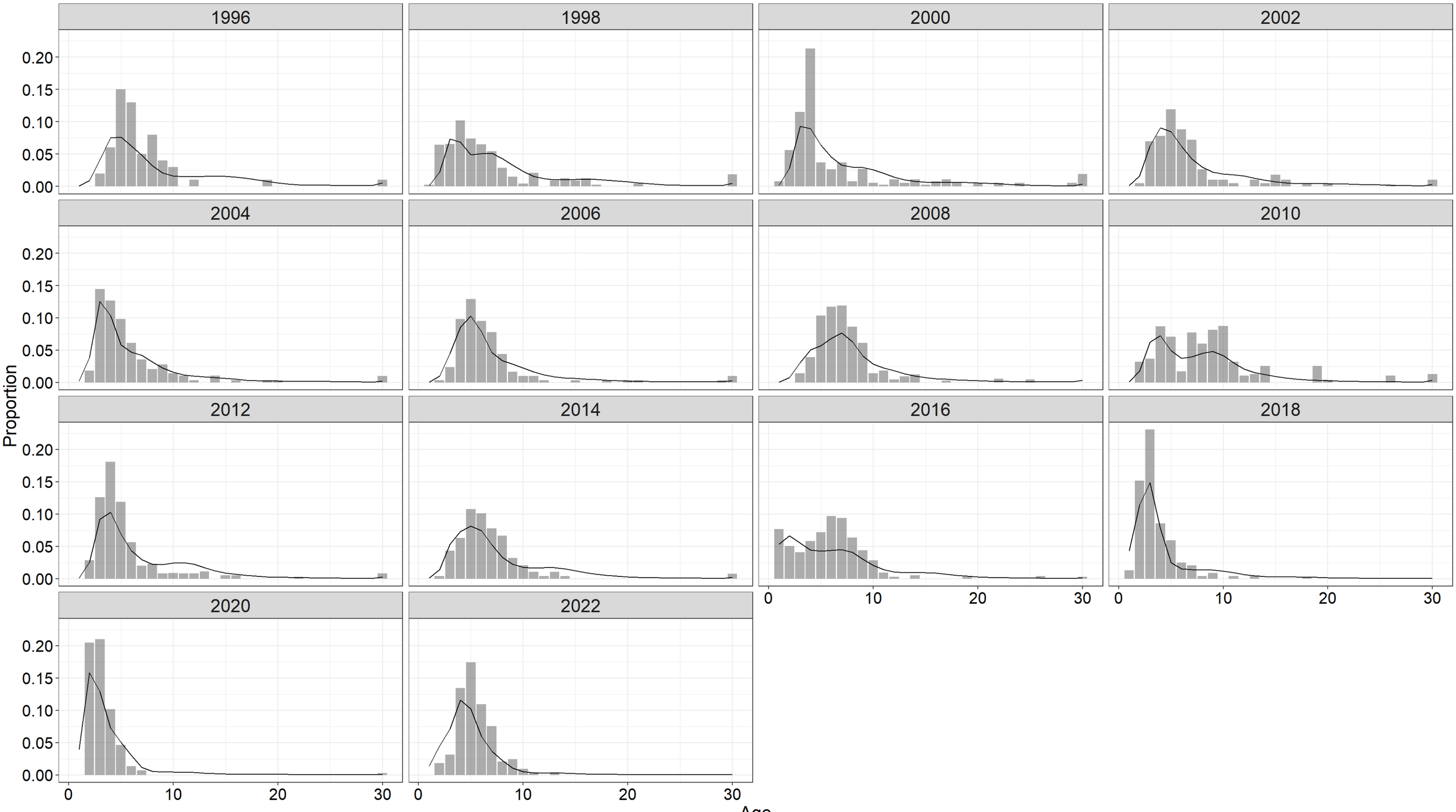
1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

Proportion
0.00 0.05 0.10 0.15
0 10 20 30
Age

BS LL Survey Age Comps OSA AK



AI LL Survey Age Comps Female AK



AI LL Survey Age Comps Male AK

1996 1998 2000 2002
2004 2006 2008 2010
2012 2014 2016 2018
2020 2022

Proportion
0.15 0.10 0.05 0.00
0 10 20 30
Age

AI LL Survey Age Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Age

30

20

10

0

1995 2000 2005 2010 2015 2020 1995 2000 2005 2010 2015 2020

Year

AK

SDNR=0.98
(0.95-1.05)

Sample quantiles

2

0

-2

-2 0 2

Theoretical quantiles

Female

Male

AK

Proportions

0.100

0.075

0.050

0.025

0.000

0 10 20 30 0 10 20 30

Age

GOA LL Survey Age Comps Female AK

1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025

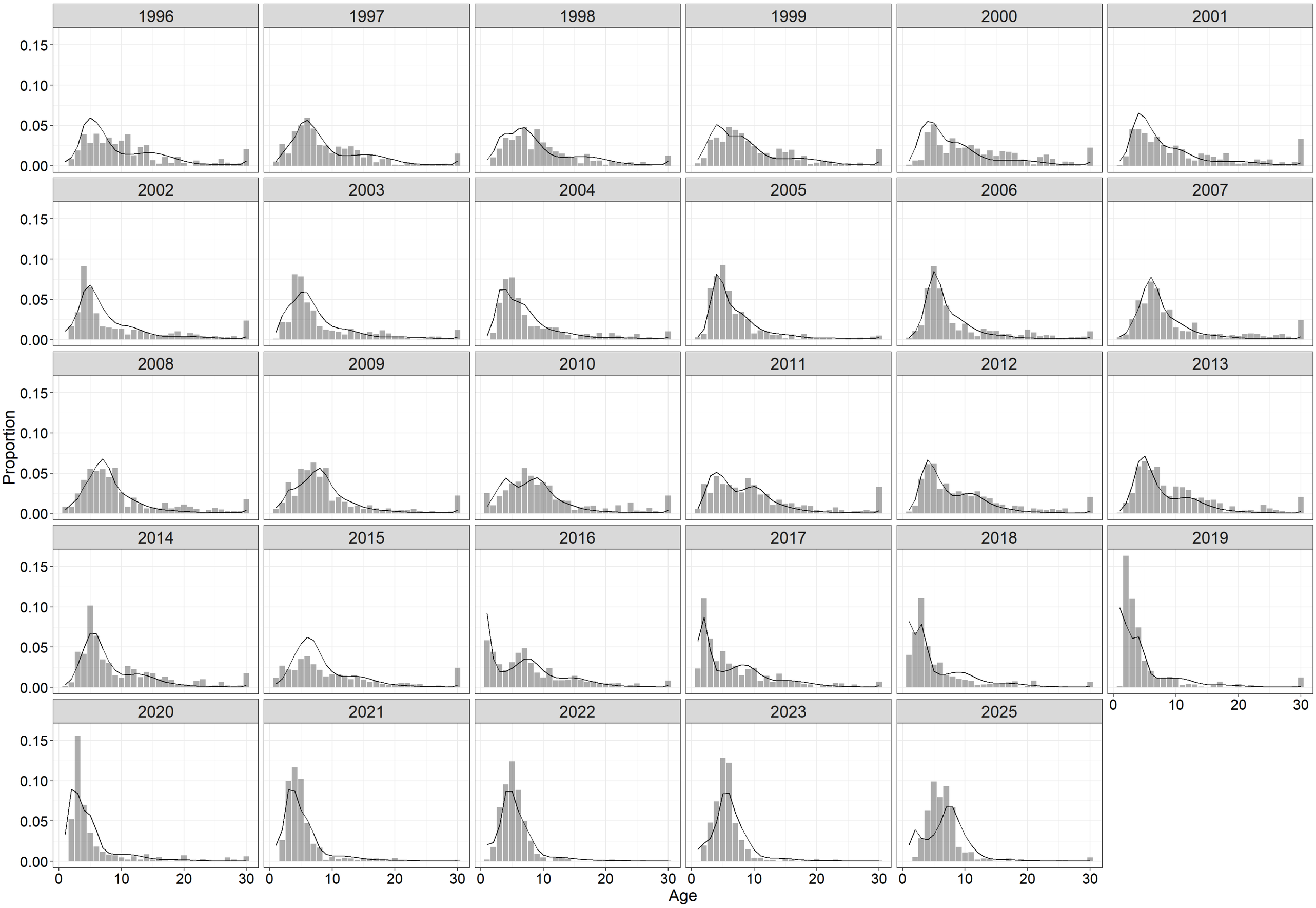
Proportion

0.0 0.1 0.2

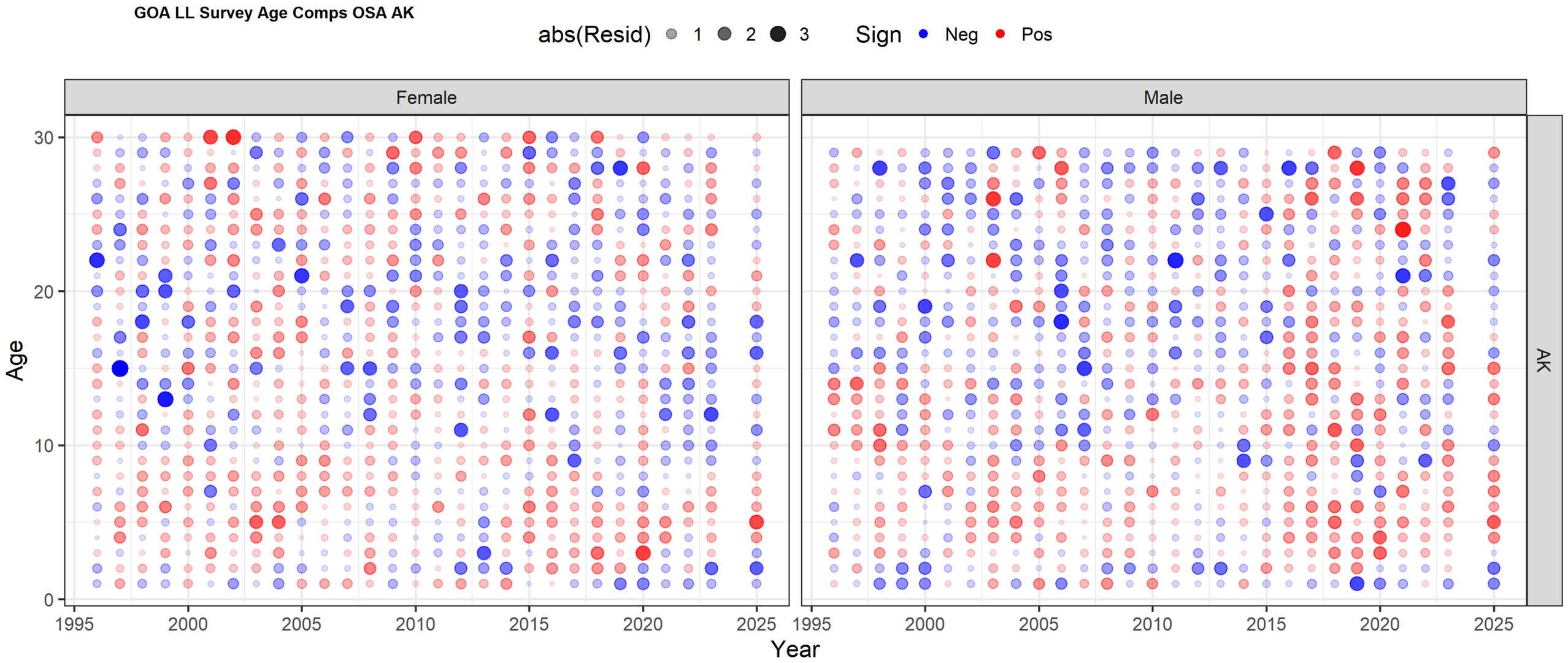
0 10 20 30

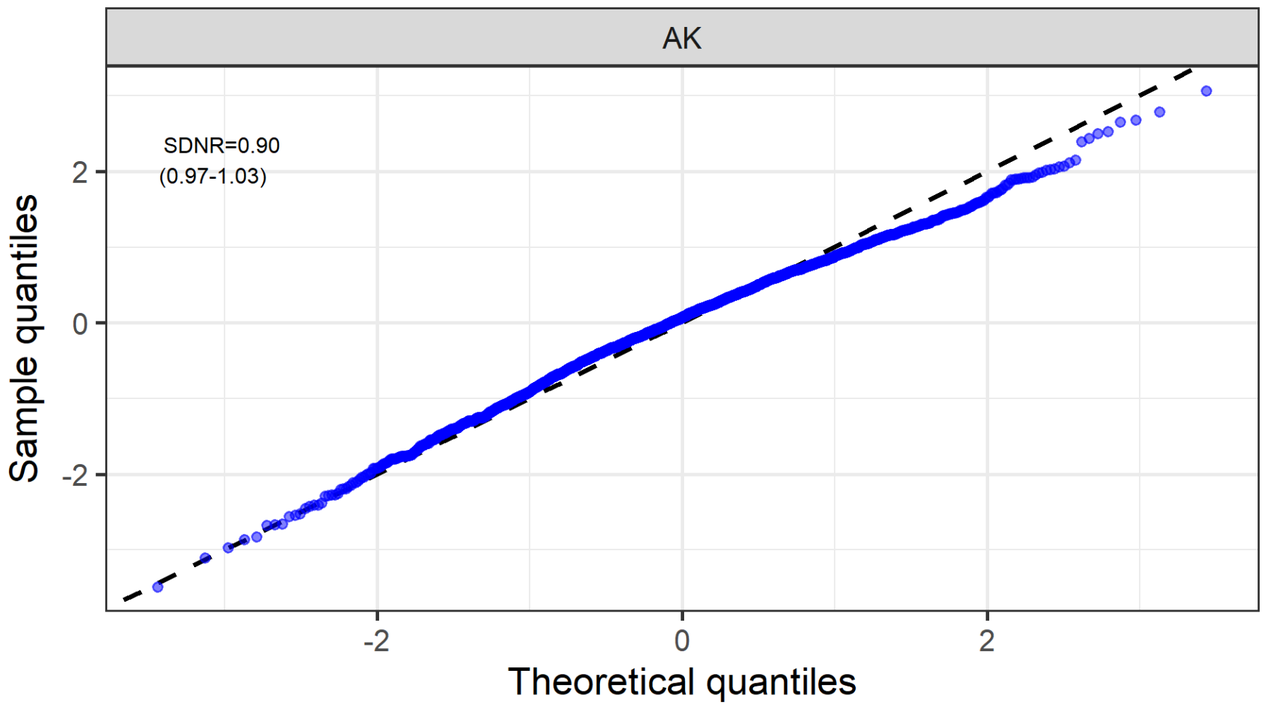
Age

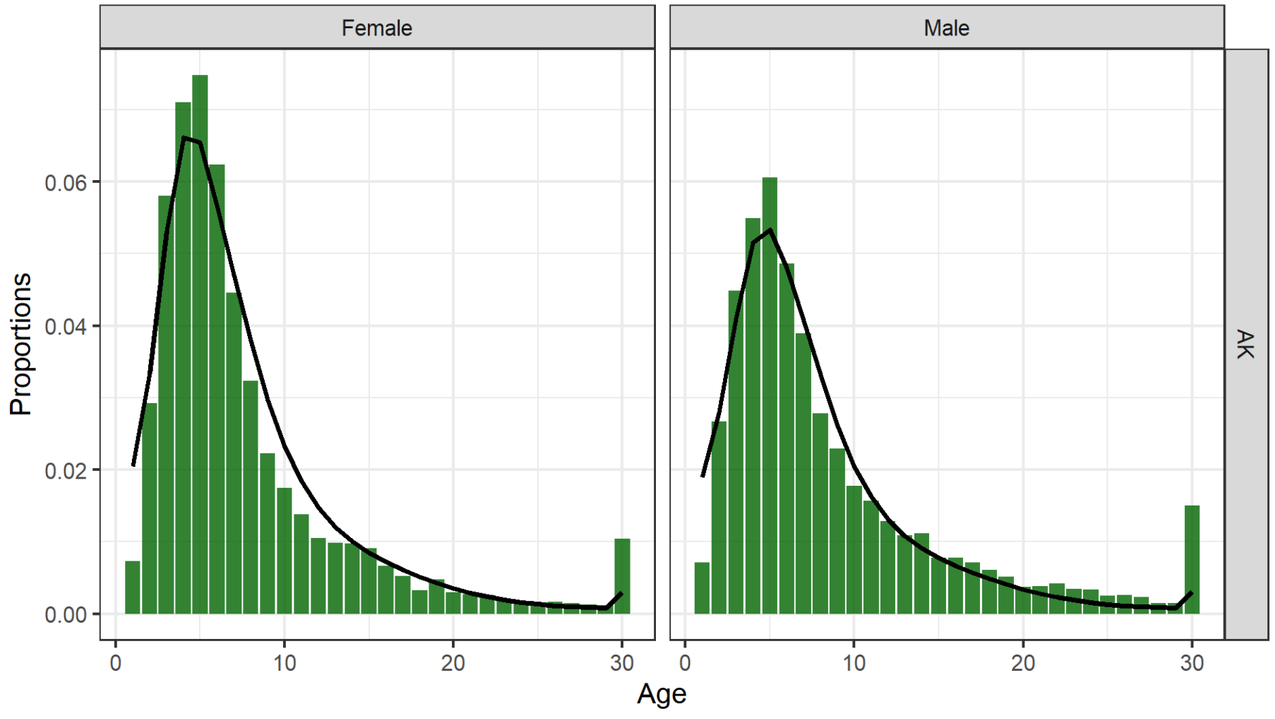
GOA LL Survey Age Comps Male AK

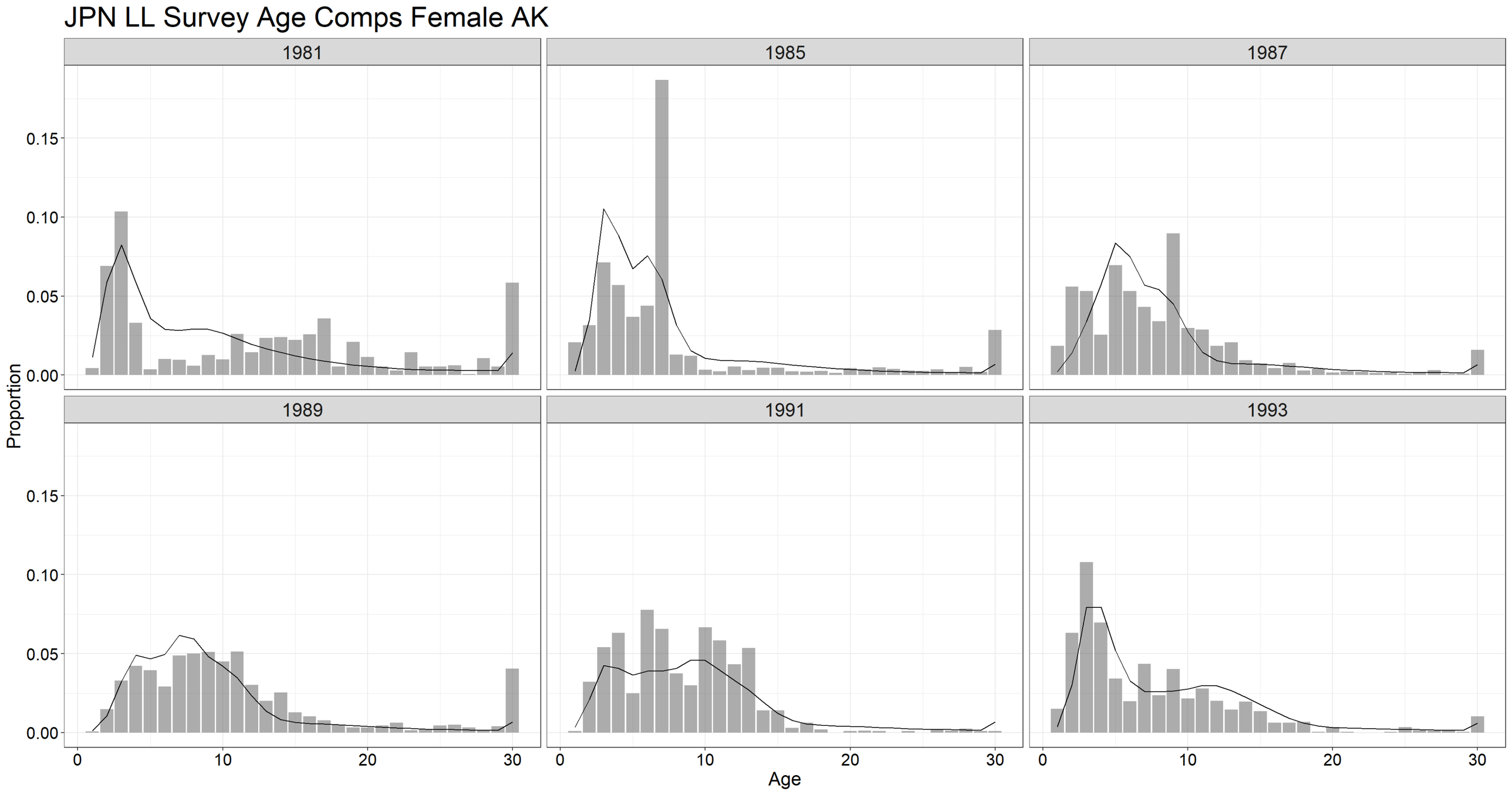


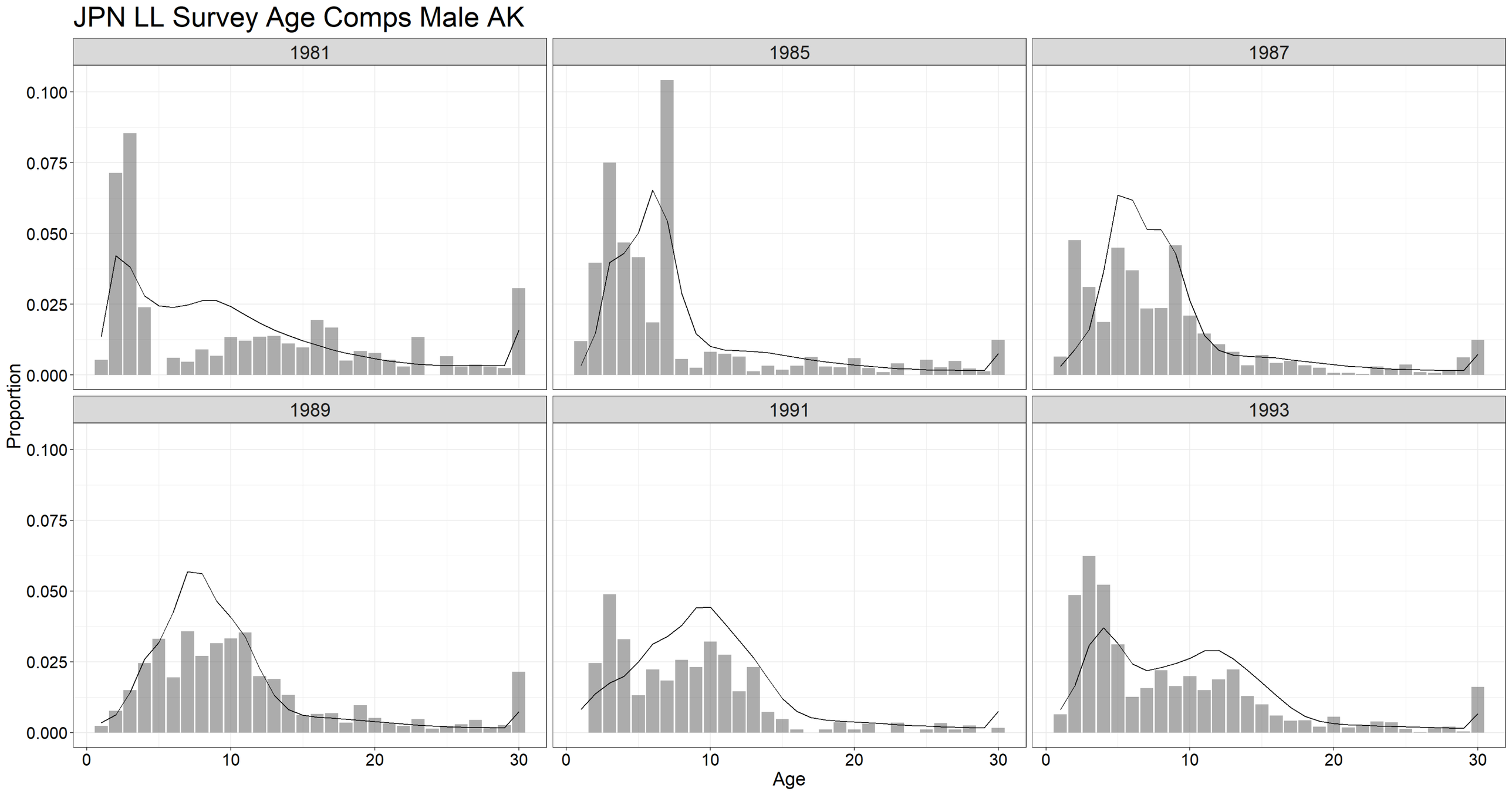
GOA LL Survey Age Comps OSA AK



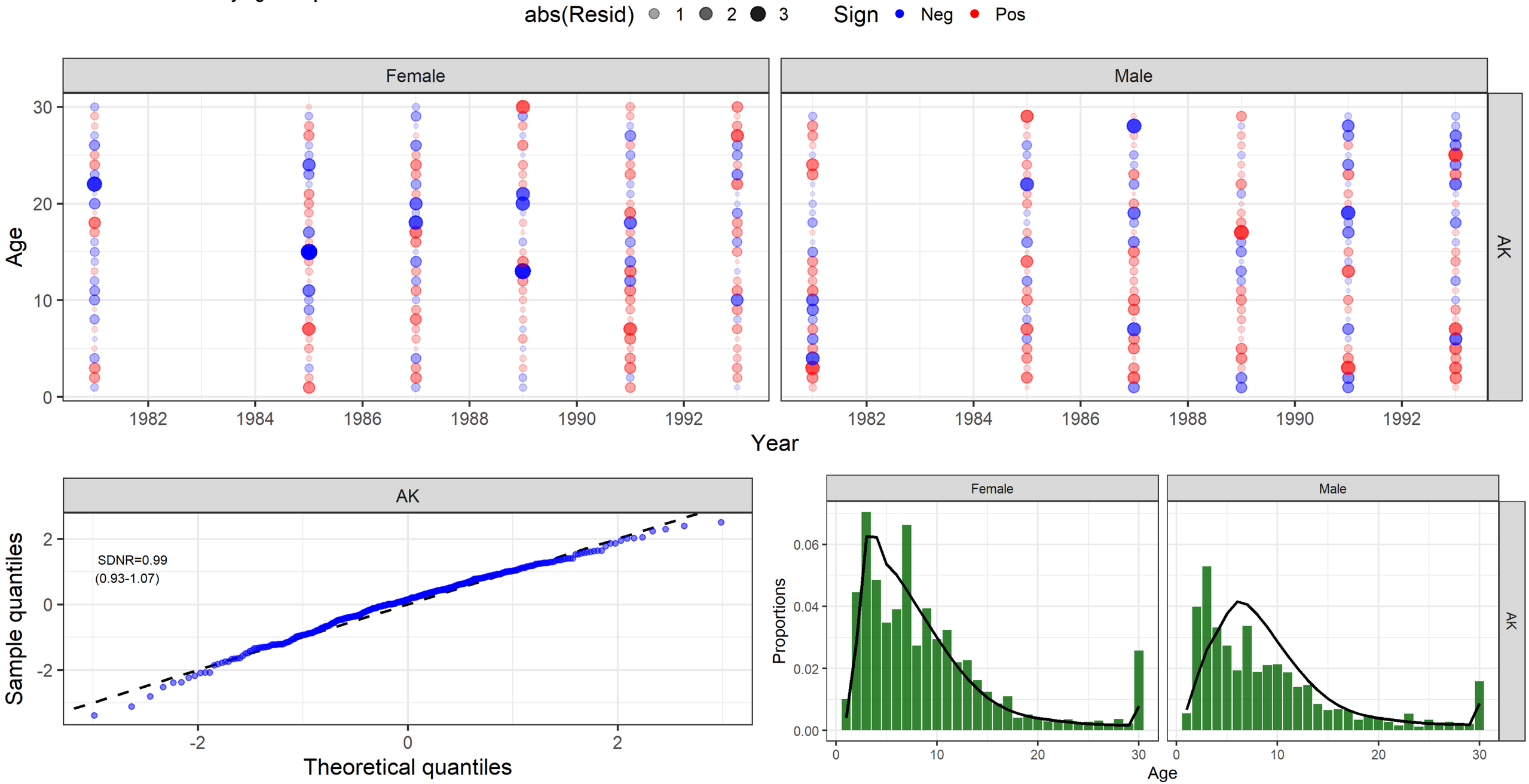





JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
Proportion
0.100
0.075
0.050
0.025
0.000
0
10
20
30
Age

JPN LL Survey Age Comps OSA AK



JPN LL Survey Length Comps Female AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

JPN LL Survey Length Comps Male AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

JPN LL Survey Length Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Len

100

80

60

40

1980 1985 1990 1980 1985 1990

Year

AK

SDNR=0.90
(0.95-1.05)

Sample quantiles

2 0 -2

-2 0 2

Theoretical quantiles

Female

Male

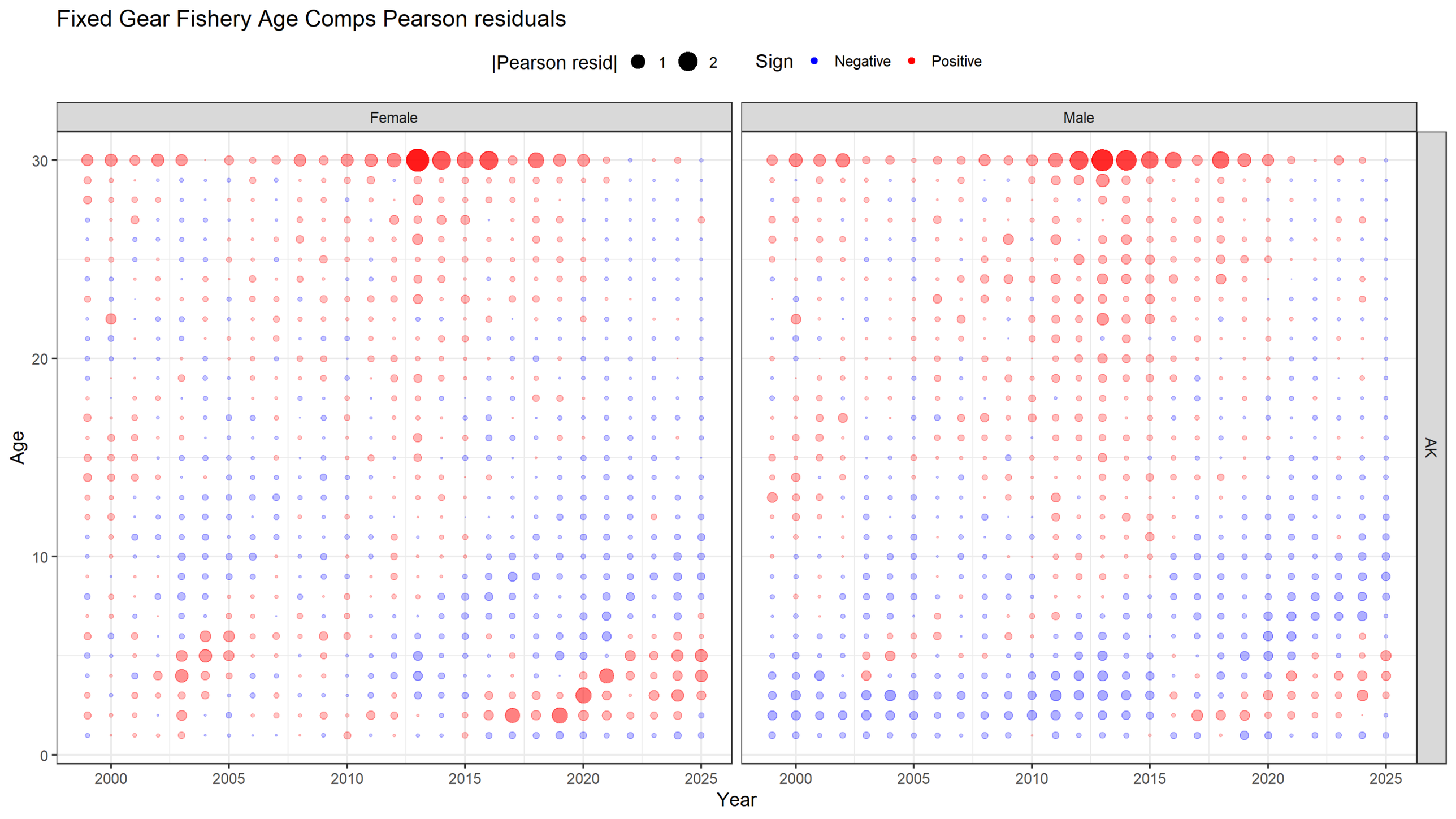
AK

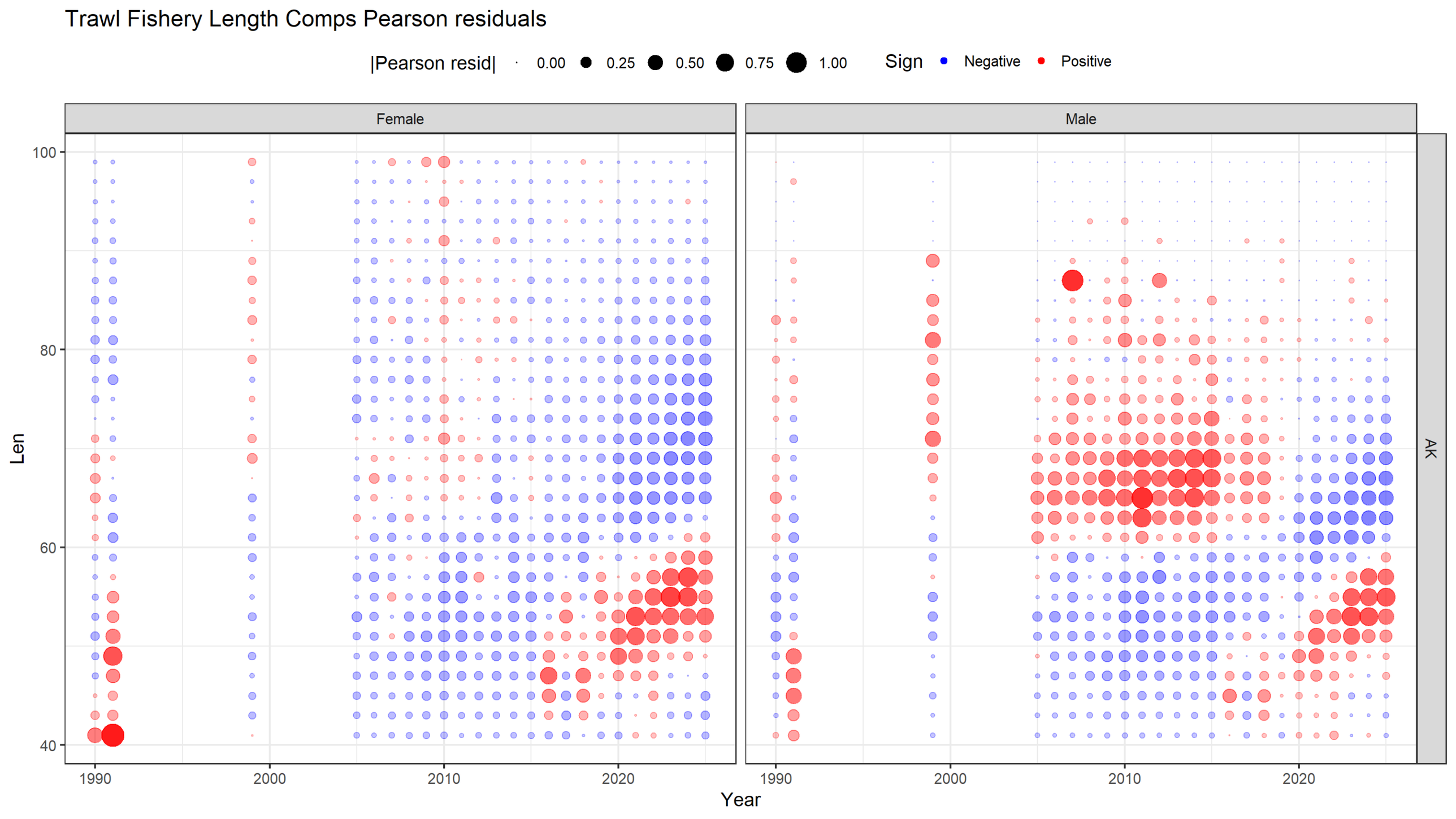
Proportions

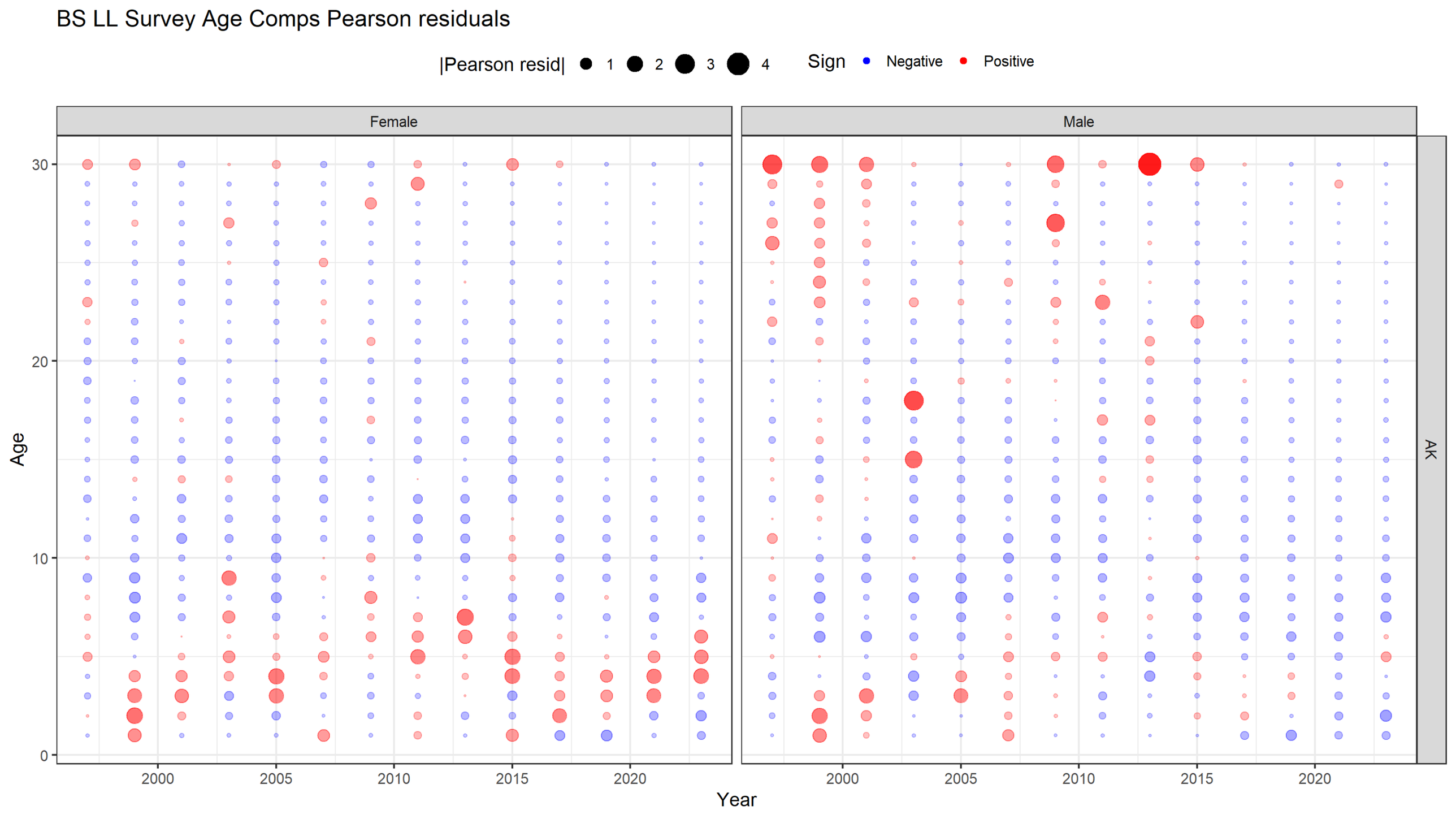
0.06 0.04 0.02 0.00

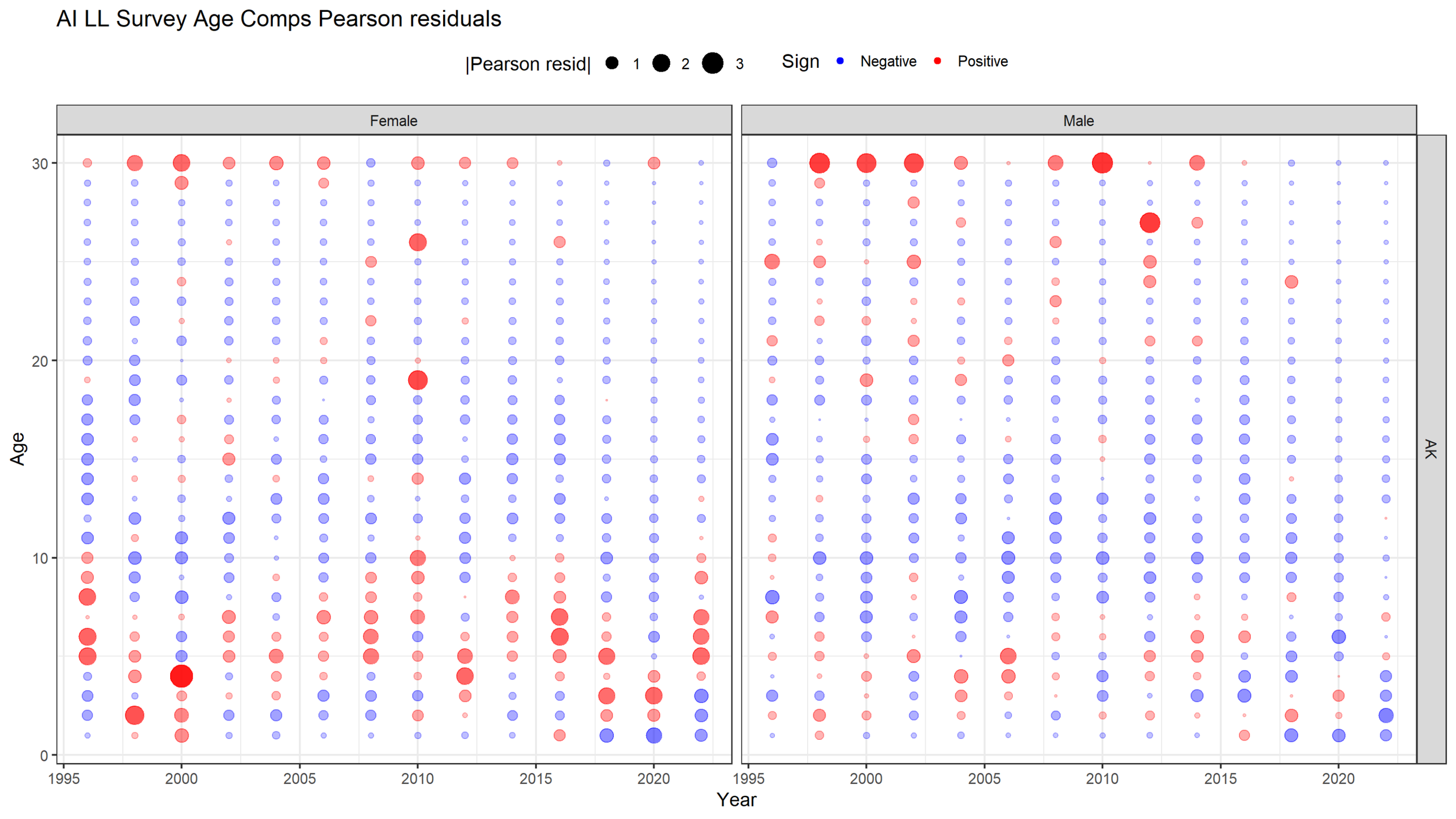
40 60 80 100 40 60 80 100

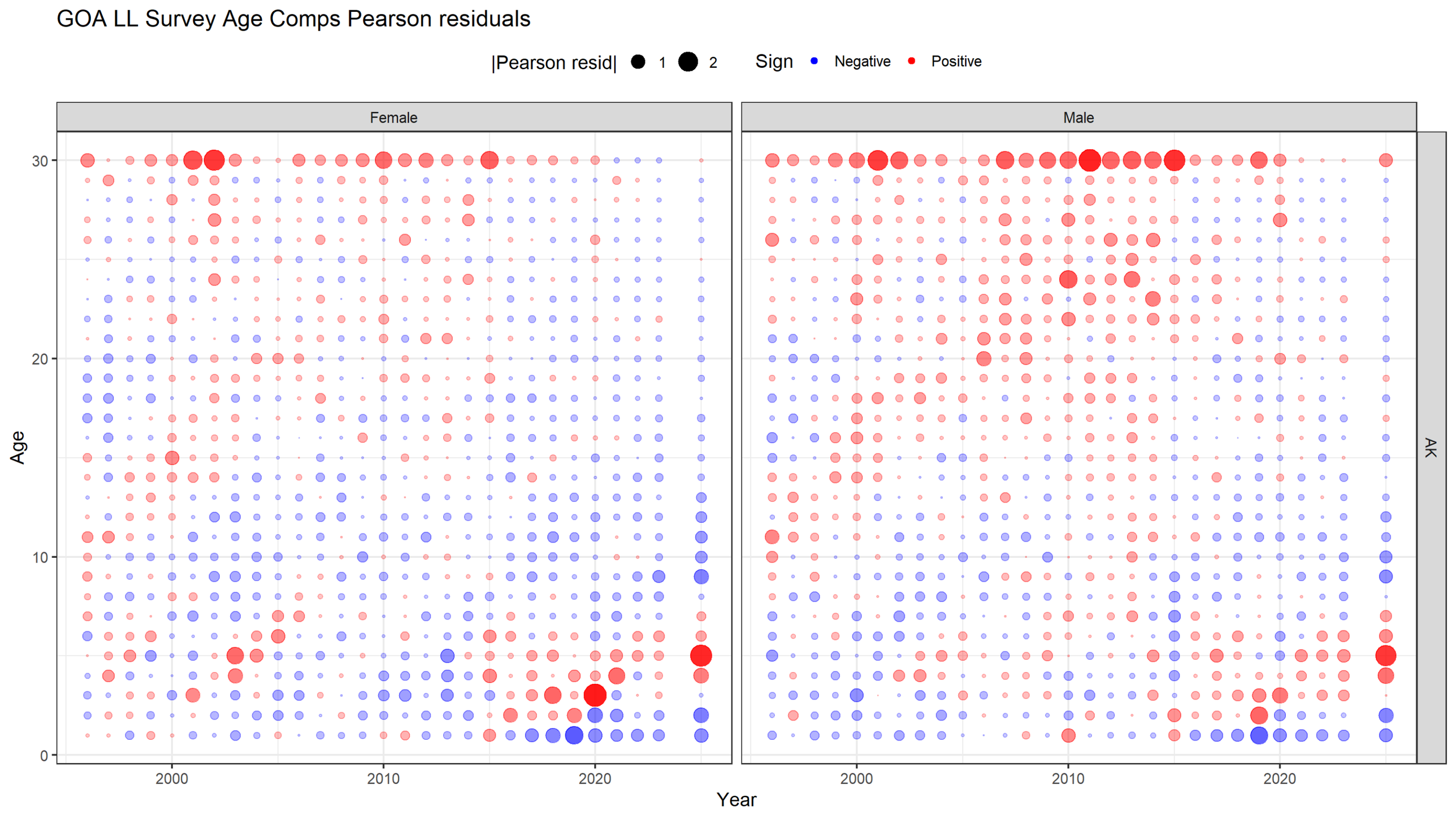
Len

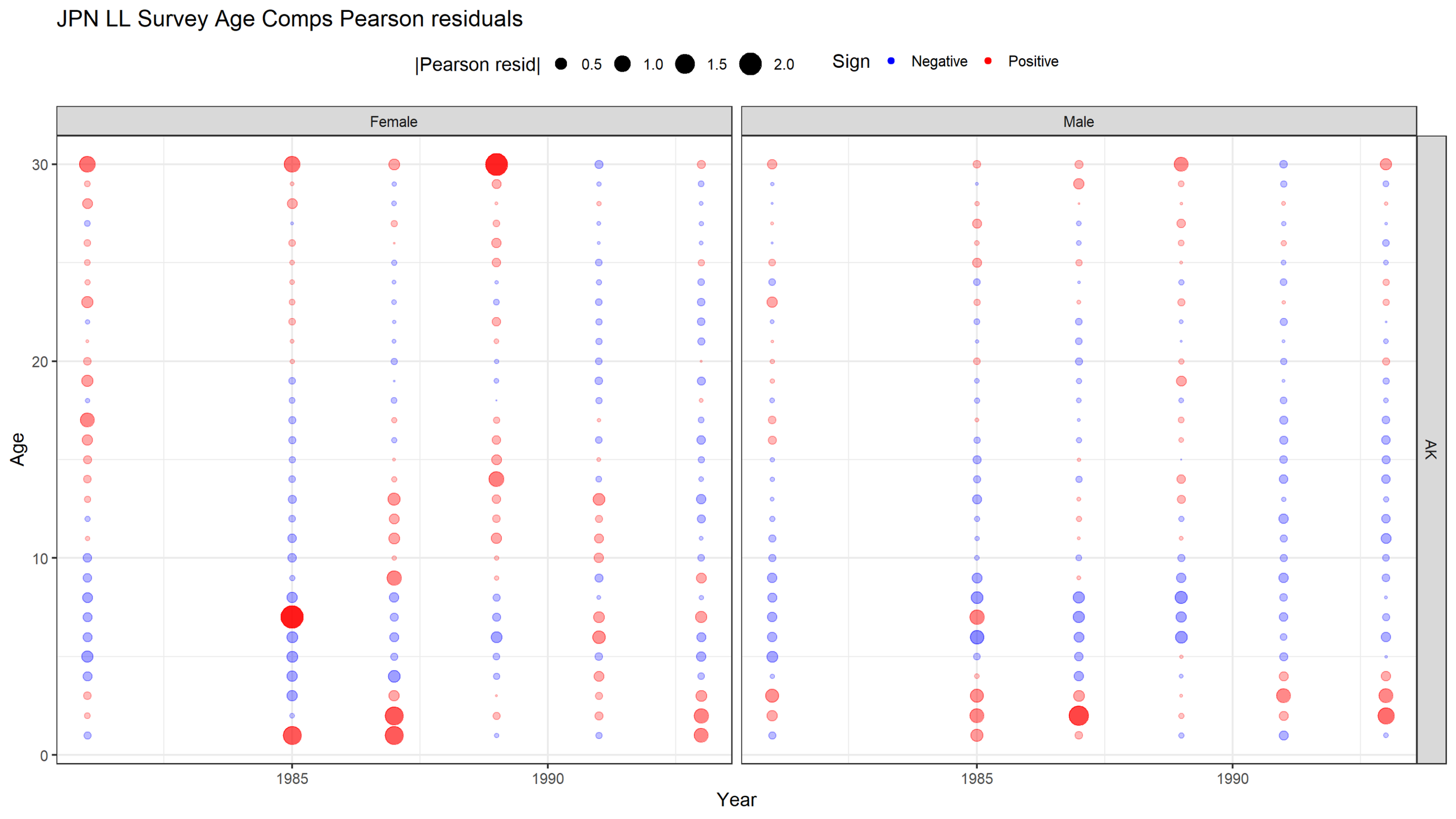

Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
1
2
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

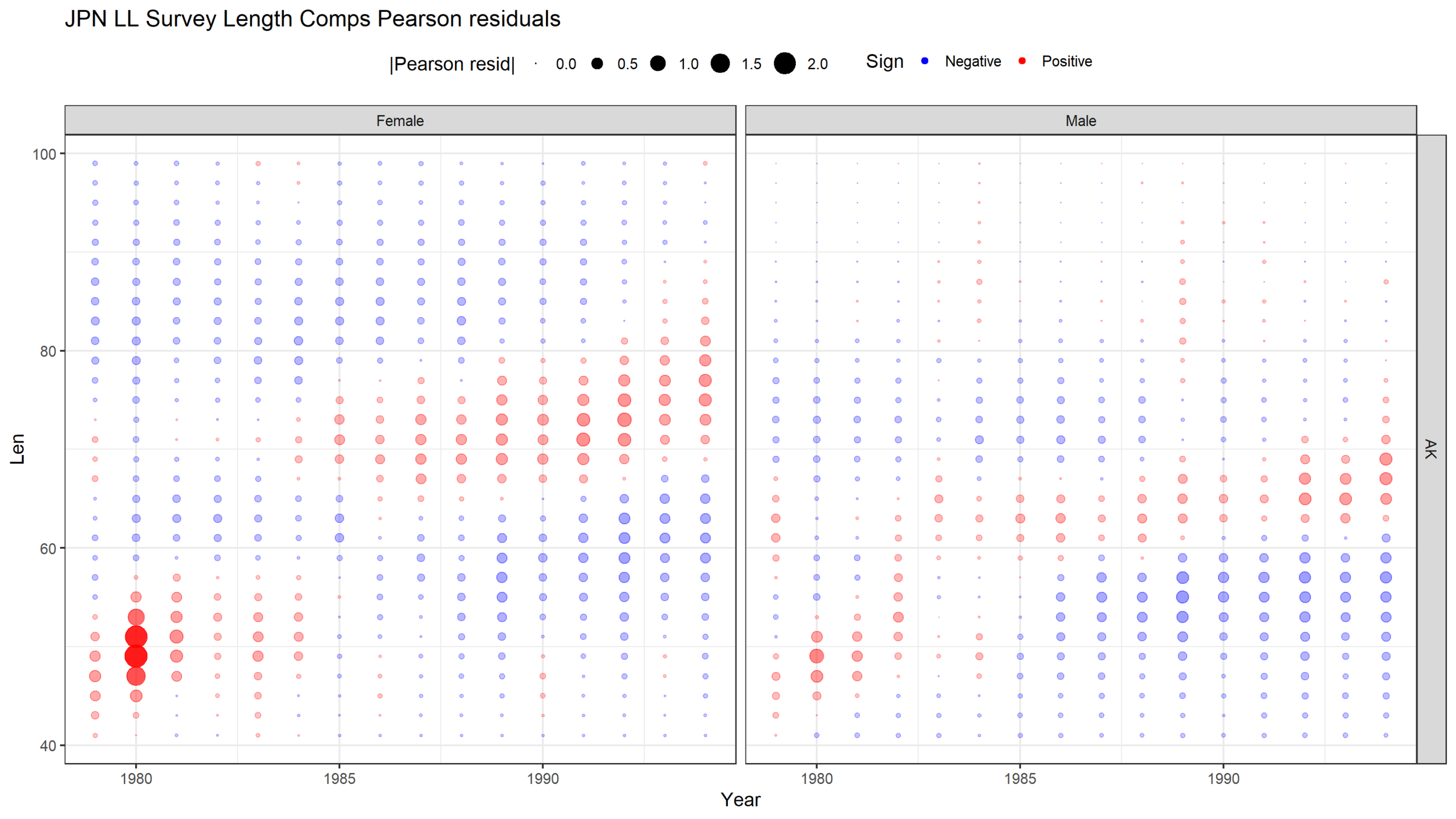

Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.00
0.25
0.50
0.75
1.00
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year


BS LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3 4
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020


AI LL Survey Age Comps Pearson residuals
|Pearson resid|
1
2
3
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
1995
2000
2005
2010
2015
2020


GOA LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020


JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
30
20
10
0
1985
1990
1985
1990
Year


JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

Source	SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	1.00	NA	NA
AI LL Survey Index	0.23	1.00	NA	NA
GOA LL Survey Index	0.08	1.00	NA	NA
Trawl Survey Index	0.21	1.00	NA	NA
JPN LL Survey Index	0.07	1.00	NA	NA
Fixed Gear Fishery CPUE	0.20	1.00	NA	NA
BS LL Survey Ages	NA	0.65	34.07	22.15
AI LL Survey Ages	NA	1.63	35.57	58.01
GOA LL Survey Ages	NA	0.42	67.17	28.23
JPN LL Survey Ages	NA	0.18	83.33	15.39
JPN LL Survey Lengths	NA	0.33	54.62	17.79
Fixed Gear Fishery Ages	NA	0.04	142.85	5.68
Trawl Fishery Lengths	NA	0.10	47.62	4.81

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
26.2_FAA_TS_CPUE	1	142.5183	315.3915	0.0835777	27.03454	117.5544	0.08672	1.21236	0.45188	0.96378

Phase Plane Assuming B40 from ~2018

